

THE STARRY MESSENGER

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The Shadow of Extra Dimensions: Can Kaluza-Klein
Geometry Mimic Dark Matter?

Fly-By-Wire Systems

Black-Hole No-Hair, Hawking Radiation, and
Unitarity: Investigating the
Information Paradox

the 4th annual

RMP

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Welcome to the fifth Issue of “The Starry Messenger”, the triannual science magazine of the Global Society of Young Physicists (GSYP). I am truly excited to introduce the incredible work of our students from the recently concluded *Research Mentorship Programme* (RMP) 2026. This issue features three outstanding research reports that focus on cosmology and engineering:

- 1. The Shadow of Extra Dimensions: Can Kaluza–Klein Geometry Mimic Dark Matter?**
- 2. Black-Hole No-Hair, Hawking Radiation, and Unitarity: Investigating the Information Paradox**
- 3. Fly-By-Wire Systems**

The GSYP was founded with a vision to connect young physics enthusiasts across the globe, fostering a community dedicated to advancing physics education. GSYP is a registered community interest company (C.I.C) in the UK that supports budding physicists through a range of initiatives, including the RMP, where students are paired with experienced mentors to undertake challenging research projects in physics.

After five intense weeks, we concluded the RMP 2026, marking an incredible journey for all involved. Throughout the programme, participants demonstrated exceptional dedication and curiosity, exploring complex topics such as the Friedmann equations, harmonic oscillators, and the Eddington–Finkelstein coordinates. Their efforts have culminated in the remarkable research reports we present in this issue.

Where It All Began...

The foundation of GSYP and the RMP was established through a personal experience.

Years ago, at a summer school on Quantum Cryptography (QCSYS) at the University of Waterloo, Canada, I met like-minded individuals: My dear friends and colleagues Leo Nagel and Samuel Siojo (who are the Senior Advisors at GSYP), and Raditya Saraf.

Taking the initial steps in founding GSYP, Leo and I – with other friends we made across the globe thanks to QCSYS – embarked on a research project titled “Measurement of Thermal Conductivity of Thin and Thick Layer Graphene via Phonon Dispersion by Inelastic X-Ray Collision Produced Through



Bremsstrahlung Radiation” for the CERN Beamline for Schools competition.

Our team spanned four continents but, despite reaching out to several professors, we struggled to find a mentor willing to guide such a geographically scattered team. Thus, we took on the challenge ourselves, learning and working together across borders and time zones. We finished our paper but, for me, the experience of working without a mentor underscored the importance of guided support for students undertaking research projects, laying the groundwork for what would become GSYF and the RMP. This experience then inspired the creation of GSYF and the RMP, which aims to provide young physics enthusiasts with the mentorship and resources we lacked.

Today, I am immensely proud of the outstanding work our students and mentors have accomplished throughout the four years since RMP was first founded.

Acknowledgements

I would like to extend my thanks to the distinguished mentors of RMP 2026 and now senior fellows of GSYF:

Kajol Mistry, who completed her MSci at King’s College London, has led the Mathematics team last year and the

Engineering team this year. She has remained an invaluable part of RMP for the past two years. Kajol has always been a wonderful friend, and an equally wonderful colleague. Her lectures on foundational mathematics have consistently given students the grounding they need to venture into more advanced ideas and pursue their own research with confidence.

The influence of her teaching can be seen throughout the work of our participants, particularly in the engineering paper, where many of the core mathematical concepts she introduced were put into practice. Her ability to lay strong foundations while opening the door to deeper exploration has been invaluable. I am immensely grateful to Kajol for generously sharing her knowledge and mathematical expertise with the students of RMP.

Nikita Azevedo, a master’s student at the University of the Witwatersrand, has also stood by us for the past two years. Nikita’s lecture notes have been a source of inspiration to us all, and the thought and care she has poured into her teaching have been nothing short of remarkable.

Perhaps the greatest testament to Nikita’s expertise is that she taught general relativity to high-school students. To take a subject of such extraordinary mathematical and



conceptual depth and make it accessible to students at this stage of their education requires not only an exceptional command of the subject, but also a rare ability to communicate difficult ideas with clarity and intuition. Her willingness to introduce our students to concepts far beyond the usual school curriculum embodies exactly the kind of intellectual ambition that RMP hopes to foster.

On a more personal note, I am incredibly grateful that GSYF gave me the opportunity to meet a great mind like hers. With an entire continent between us, our paths might otherwise never have crossed. The fact that this organisation can bring together people, ideas, and teams across such distances makes me especially proud. In moments like these, I can see its purpose being fulfilled in the most meaningful way.

Tobias Bhanji, a fourth-year MSci student at King's College London, has been with us since the second annual RMP – three years of dedication and contribution that have never gone unnoticed.

I still remember the first time I met Toby. What immediately stood out was the depth of his understanding of cosmology, but over the years, I have come to appreciate just as much the rigour he brings to his lectures and the confidence with which he guides his

students. Whether explaining challenging cosmological concepts, or stepping up to lead his team when needed, Toby has consistently brought both expertise and commitment to RMP. His contribution over these past three years has left a lasting mark, and I am deeply grateful for everything he has given to the programme.

I would also like to extend my heartfelt thanks to our wonderful team, whose work behind the scenes has been every bit as important in bringing RMP to life:

Paul San Diego, a Computer Science graduate from King's College London, generously volunteered his time and expertise to design, build, and manage our website. Much of what we have been able to share with the wider world has been made possible through his work, keeping the website user-friendly and accessible with efficient design. I am incredibly grateful for the time, care, and creativity he has invested in GSYF.

I would also like to give a very special thank you to *Hayley Snelling*, a Physics MSc graduate from University College London, whom I had the pleasure of meeting through GSYF and who served as the editor for this issue. It has been a genuine pleasure working with her and, once again, I find myself grateful for the remarkable people and



collaborations that GSYF has brought into my life. Her contribution has helped turn a collection of ideas and papers into an issue we can all be proud of.

Last but certainly not least, I would like to thank *our students*, now proudly RMP Alumni, without whom none of this would have been possible. Your curiosity, determination, and willingness to challenge yourselves are what gave this programme its purpose. Every lecture attended, every question asked, every difficult problem wrestled with, and every paper brought to completion is a testament to the hard work you put into this experience.

It has been a privilege to watch you grow as students, researchers, and thinkers. While your time as RMP participants may have come to an end, I hope the curiosity and ambition that brought you here will stay with you wherever you go next. Thank you for giving this programme your time and energy and, above all, for making RMP what it is. Welcome to your community.

Join Us!

As always, we at GSYF encourage you to support and join our mission to further physics education and connect young physics enthusiasts worldwide. For more information, please refer to our posters at the end of our magazine.

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Become a Member of GSYF!

As a member of GSYF, you gain access to exclusive benefits that will enhance your journey as a young physicist. Members can request reviews from research mentors who are conducting cutting-edge studies in physics. After a thorough editorial review, you can submit your work for publication in our science magazine. Additionally, you will have the opportunity to attend talks by renowned science experts brought in by GSYF.

Membership is currently free, so join us and take advantage of these incredible opportunities.

Science Magazine Schedule

Our Science Magazine will be published three times a year:

- ❖ **April:** Independent research published before the annual RMP
- ❖ **August:** Research reports from the annual RMP
- ❖ **December:** Independent research published after the annual RMP

Stay tuned for our next issue in December 2026, where we will feature more research reports and provide details on the upcoming RMP 2027!

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We invite you to become a GSYF member and take the first step towards your research dreams. Access expert talks, publish your research, and connect with a global community of young physicists. Don't miss out on this incredible opportunity to unlock your potential as a physics researcher.

*The views of our writers are their own.
We aim for sound science but not editorial orthodoxy.*

The Shadow of Extra Dimensions: Can Kaluza–Klein Geometry Mimic Dark Matter?

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Abstract

Dark matter is one of the major unresolved problems in modern physics, with evidence from galaxy rotation curves, gravitational lensing, and cosmological observations indicating more gravitational mass than can be accounted for by visible matter. This paper investigates whether this missing gravitational effect could instead arise from higher-dimensional geometry. Using a simplified Kaluza–Klein framework, we compactify a fifth spatial dimension and derive the resulting four-dimensional effective theory. The size of the extra dimension appears as a scalar field, known as the radion, whose stress-energy contributes to the gravitational field. Then we study the cosmological evolution of an oscillating radion and see whether its energy density can scale as $\rho_\sigma \propto a^{-3}$, the characteristic behaviour of cold dark matter at the level of homogeneous cosmological background.

1 Introduction

The Λ CDM model provides an extremely successful description of the large-scale evolution of the universe, but its success depends on the existence of a substantial dark matter component whose physical nature remains unknown [1]. The existence of dark matter is inferred from the gravitational effects required to explain observations ranging from galaxy rotation curves and gravitational lensing to the cosmic microwave background and the growth of large-scale structure. Thus, while the gravitational evidence for an additional matter-like component is extensive, the underlying origin of this component remains an open problem. This therefore opens a line of inquiry in cosmology: does the additional gravitational effect attributed to dark matter necessarily have to arise from a new form of matter? If the effective four-dimensional description of the universe can contain fields generated by the geometry of a higher-dimensional space-time, then some phenomena normally interpreted as evidence for additional matter could instead have a geometric origin. This provides a direct motivation for considering theories in which extra dimensions produce observable four-dimensional degrees of freedom.

Kaluza–Klein theory is a particularly simple framework in which to investigate this possibility. When a fifth spatial dimension is compactified, the higher-dimensional metric decomposes into familiar four-dimensional gravity and gauge forces together with a scalar field, called the radion, descended from the geometry of the compact dimension. This paper investigates whether the radion could, in principle, acquire an effective stress-energy tensor whose cosmological behaviour resembles that of cold dark matter. We derive the four-dimensional effective theory produced by Kaluza–Klein compactification, identify and canonically normalise the resulting radion field, and investigate its cosmological dynamics. In particular, we ask whether a stabilised radion undergoing coherent oscillations can behave as an approximately pressureless component with $\rho_\sigma \propto a^{-3}$, as required for dark matter. The aim is not to claim that the Kaluza–Klein theory has already explained dark matter, but to test whether extra-dimensional geometry can reproduce one of the defining dynamical properties of dark matter.

2 Evidence and Requirements for Dark Matter

The existence of dark matter was first inferred by discrepancies between the observed gravitational behaviour of galaxies and the predictions obtained from their visible matter content. If a galaxy consists only of observable stars, gas, and dust, then Newtonian gravity predicts a specific relationship between orbital velocity and distance from the galactic centre.

For a star of mass m orbiting a galaxy at radius r , the gravitational force is

$$F_g = \frac{GM(r)m}{r^2}, \quad (1)$$

where G is Newton's gravitational constant, $M(r)$ is the total mass enclosed within radius r , and m is the mass of the orbiting star. For a stable circular orbit, the gravitational force provides the centripetal force

$$F_c = \frac{mv(r)^2}{r}, \quad (2)$$

with $v(r)$ being the linear velocity of the orbiting party. Equating these two expressions gives:

$$\frac{GM(r)m}{r^2} = \frac{mv(r)^2}{r}. \quad (3)$$

Cancelling m and rearranging leads to:

$$v(r)^2 = \frac{GM(r)}{r}. \quad (4)$$

This equation predicts the expected rotation curve of a galaxy. If most of the mass is concentrated near the galactic centre, then at large distances the enclosed visible mass approaches a constant value:

$$M(r) \approx M_{\text{visible}}. \quad (5)$$

Therefore,

$$v(r)^2 \approx \frac{GM_{\text{visible}}}{r}, \quad (6)$$

$$v(r) \propto r^{-1/2}. \quad (7)$$

This means that the orbital velocity of stars should decrease as their distance from the galactic centre increases. However, observations show a very different behaviour [2]: instead of decreasing, the velocity remains approximately constant

$$v(r) \approx v_0. \quad (8)$$

This implies

$$v_0^2 = \frac{GM(r)}{r}, \quad (9)$$

such that

$$M(r) = \frac{v_0^2 r}{G}, \quad (10)$$

which means

$$M(r) \propto r. \quad (11)$$

The amount of gravitational mass enclosed within the galaxy bizarrely continues increasing approximately linearly with radius, even though the visible matter distribution does not. This implies that galaxies contain a large quantity of additional unseen mass. According to general relativity, massive objects distort the geometry of space-time. Light follows geodesics within this curved spacetime, meaning that the path of light from distant objects is altered by gravitational fields [3]. The deflection angle of light passing near a mass M is approximately

$$\alpha = \frac{4GM}{c^2 b}, \quad (12)$$

where c is the speed of light, M is the lensing mass, and b is the impact parameter: the perpendicular separation between the deflecting mass M and the geodesic that the light would have followed if left undisturbed. Measuring the bending of light allows the gravitational mass of an object to be inferred [4].

Observations of galaxy clusters show that the gravitational mass determined from lensing is significantly larger than the mass calculated from visible matter alone. The lensing signal behaves as if an additional invisible mass distribution surrounds galaxies and clusters [5]. One of the strongest pieces of evidence that we have for dark matter comes from observations of colliding galaxy clusters, particularly the system known as the Bullet Cluster [6]. During a galaxy cluster collision, the various components of the cluster behave differently. The hot gas containing most of the ordinary matter interacts electromagnetically and slows down. The galaxies themselves, containing relatively little mass compared with the gas, pass through each other with minimal interaction. Gravitational lensing reveals that the majority of the gravitational mass lies away from the hot gas. This separation implies that the dominant gravitational component does not behave like ordinary matter. If gravity were produced only by visible matter, the strongest gravitational field should coincide with the location of the hot gas. Instead, the gravitational potential is displaced, implying the presence of dark matter. Dark matter is also required by observations of the universe on the largest scales. The evolution of the universe is described by the Friedmann equation:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{\kappa}{a^2} + \frac{\Lambda}{3}, \quad (13)$$

where H is the expansion rate, ρ is the total energy density, κ represents spatial curvature, Λ is the cosmological constant, and $a(t)$ is the cosmic scale factor. The total matter (both baryonic and dark) density affects the formation of galaxies, the growth of cosmic structures, and the anisotropies of the cosmic microwave background. Measurements of the cosmic microwave background show that ordinary baryonic matter alone cannot produce the observed pattern of structure formation [7]. The observed universe requires an additional non-luminous matter component [1].

In the Λ CDM model:

$$\rho_{\text{total}} = \rho_b + \rho_{DM} + \rho_{\Lambda} + \rho_r, \quad (14)$$

where:

- ρ_b is baryonic matter density
- ρ_{DM} is dark matter density
- ρ_{Λ} effective energy density associated with the cosmological constant

- ρ_r is energy density of radiation

The approximate composition of the universe now is:

$$\Omega_b \approx 0.049 \quad (15)$$

$$\Omega_{DM} \approx 0.265 \quad (16)$$

$$\Omega_\Lambda \approx 0.685 \quad (17)$$

with the almost negligible remainder being radiation. This means that ordinary matter represents only a small fraction of the total cosmic energy [1]. A successful dark matter candidate must satisfy several conditions:

- It must contribute gravitationally
- It must behave approximately like pressureless matter
- Its density must decrease as the universe expands:

$$\rho_{DM} \propto a^{-3} \quad (18)$$

where $a(t)$ is the scale factor (derived in this paper using the Klein-Gordon Equation) [8].

3 The Kaluza–Klein Framework

The standard interpretation introduces a new particle species to explain dark matter. However, another possibility is that the missing gravitational effects originate from the structure of spacetime itself. Higher-dimensional theories provide a natural framework for investigating this idea [9][10][11]. In Kaluza–Klein theory, the universe is assumed to contain additional compact spatial dimensions. Although these dimensions may be too small to observe directly, their geometry can influence the physics measured in four-dimensional spacetime. The purpose of this paper is to investigate whether or not hidden-dimensional geometry can generate an effective gravitational contribution that mimics dark matter. To test this possibility, this work studies the scalar fields generated by Kaluza–Klein compactification and determines whether their effective stress-energy tensor can reproduce the cosmological behaviour expected of dark matter.

Throughout this paper, indices with uppercase Latin characters (A, B) run from 0–4, Greek characters (μ, ν) run from 0–3 and lowercase Latin characters (i, j) run from 1–3.

General relativity describes gravity as the curvature of a four-dimensional spacetime manifold M^4 with coordinates

$$x^\mu = (x^0, x^1, x^2, x^3), \quad (19)$$

where $x^0 = t$ represents time, and x^1, x^2, x^3 represent the three spatial dimensions. The geometry of spacetime is described by the 4×4 metric tensor $g_{\mu\nu}$, which determines distances through the line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (20)$$

Einstein's field equations relate this geometry to matter:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (21)$$

where $G_{\mu\nu}$ is the Einstein tensor describing curvature, G is the Newtonian gravitational constant, and $T_{\mu\nu}$ is the stress-energy tensor describing matter and energy. However, one possible extension of general relativity is that the universe may contain additional spatial dimensions beyond the four dimensions we observe. The earliest successful higher-dimensional theory was proposed by Theodor Kaluza in 1921 [10] and further expounded by Oskar Klein in 1926 [11]. A five-dimensional theory of gravity, when viewed from a four-dimensional perspective, naturally produces both gravity and electromagnetism [9].

Instead of considering spacetime as M^4 , Kaluza–Klein theory considers a five-dimensional manifold M^5 with coordinates

$$x^A = (x^0, x^1, x^2, x^3, y), \quad (22)$$

where indices 0–3 represent ordinary spacetime coordinates and $y = x^4$ represents an additional spatial dimension. The metric is therefore expanded to a 5×5 tensor [12] g_{AB} with indices $A, B = 0, 1, 2, 3, 4$. The five-dimensional line element becomes:

$$ds^2 = g_{AB} dx^A dx^B, \quad (23)$$

or, if we write it explicitly,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu + 2g_{\mu 5} dx^\mu dy + g_{44} dy^2, \quad (24)$$

where μ and ν run from 0 to 3 as normal.

If an additional spatial dimension exists, it is not observed in everyday experience. Kaluza–Klein theory tries to resolve this by assuming that the extra dimension is compactified [9][10][11]. Instead of extending infinitely,

$$-\infty < y < \infty \quad (25)$$

the fifth dimension is assumed to form a “tiny” circle (still an open question in the field)

$$y \sim y + 2\pi R \quad (26)$$

where R is the radius of compactification. This means that travelling a distance $2\pi R$ along the fifth dimension returns an object to its original position.

The geometry of spacetime therefore becomes:

$$M^5 = M^4 \times S^1 \quad (27)$$

where: M^4 is the ordinary four-dimensional spacetime and S^1 is the compact fifth dimension. A useful analogy is a hosepipe viewed from far away. An observer at a far distance sees only a one-dimensional line, but a small insect travelling along the hosepipe can detect the circular direction around it. Similarly, if R is extremely small, ordinary observers may perceive only four dimensions. The original Kaluza–Klein theory assumes that most physical quantities are independent of the compact dimension. Mathematically:

$$\frac{\partial}{\partial y} = 0 \quad (28)$$

This is known as the cylindrical condition. It means that the fields appear identical at every point around the compact dimension. The five-dimensional metric then depends only on the four observable coordinates:

$$g_{AB} = g_{AB}(x^\mu) \quad (29)$$

This allows the higher-dimensional gravitational theory to be rewritten as an effective four-dimensional theory. For this paper, the biggest mathematical hurdle in Kaluza–Klein theory is to decompose the five-dimensional metric into four-dimensional fields and analyse what comes out of the reduction. The five-dimensional metric can be written as [9][12][13]:

$$g_{AB} = \begin{pmatrix} g_{\mu\nu} + \phi^2 A_\mu A_\nu & \phi^2 A_\mu \\ \phi^2 A_\nu & \phi^2 \end{pmatrix}. \quad (30)$$

Here $g_{\mu\nu}$ is the ordinary four-dimensional spacetime metric, A_μ is a four-dimensional gauge vector field, and ϕ is a scalar field. The corresponding line element becomes

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu + \phi^2 (dy + A_\mu dx^\mu)^2, \quad (31)$$

expanding to:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu + \phi^2 dy^2 + 2\phi^2 A_\mu dx^\mu dy + \phi^2 A_\mu A_\nu dx^\mu dx^\nu. \quad (32)$$

The off-diagonal components, $g_{\mu 4}$, of the five-dimensional metric can be identified with a four-dimensional vector field A_μ . The associated field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (33)$$

which has the same mathematical structure as the electromagnetic field [9]. Thus, in simple terms, higher-dimensional gravity in this framework breaks into gravity and electromagnetism. The component g_{44} is a scalar that describes the size of the compact dimension:

$$g_{44} = \phi^2. \quad (34)$$

Here, $\phi(x)$ is a field known as the **radion field** [14].

Unlike ordinary matter fields, the radion originates solely from geometry. This makes it particularly interesting for dark matter models. If the size of the extra dimension varies across spacetime, then $\phi = \phi(x)$ depends on position, and its energy contributes to the four-dimensional gravitational equations. In this paper, it is postulated that this geometrically generated scalar field may produce (in a simple model) an effective stress-energy contribution:

$$G_{\mu\nu} = 8\pi G(T_{\mu\nu}^{\text{visible}} + T_{\mu\nu}^\phi), \quad (35)$$

where $T_{\mu\nu}^\phi$ is not ordinary matter, but an apparent gravitational effect caused by hidden-dimensional geometry. The Kaluza–Klein framework divides the five-dimensional gravitational field into three four-dimensional components: 4D gravity, a gauge field, and a scalar (radion) field. The scalar field is of particular importance because it can influence spacetime curvature similarly to a matter source.

4 Dimensional Reduction of the Einstein–Hilbert Action

The central idea of Kaluza–Klein theory is that a higher-dimensional gravitational theory can appear, to a lower-dimensional observer, as ordinary gravity plus additional fields.

Returning to five-dimensional spacetime,

$$M^5 = M^4 \times S^1, \quad (36)$$

the fundamental assumption is that the true gravitational theory exists in five dimensions. Therefore, the starting point is not the four-dimensional Einstein equation, but the five-dimensional Einstein–Hilbert action. The action for gravity in D dimensions is [12]

$$S_D = \frac{1}{16\pi G_D} \int d^D x \sqrt{-g_D} R_D, \quad (37)$$

where G_D is the D -dimensional gravitational constant, g_D is the determinant of the D -dimensional metric, and R_D is the D -dimensional Ricci scalar. Thus, for five dimensions:

$$S_5 = \frac{1}{16\pi G_5} \int d^5 x \sqrt{-g_5} R_5. \quad (38)$$

The process of converting the higher-dimensional theory into a lower-dimensional effective theory is called dimensional reduction.

Recall the Kaluza–Klein metric [9][13][12]

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu + \phi^2 (dy + A_\mu dx^\mu)^2. \quad (39)$$

For the purpose of studying dark matter, we are mainly interested in the radion field, ϕ . To simplify the derivation, we set

$$A_\mu = 0. \quad (40)$$

This removes the electromagnetic component and allows us to focus entirely on the effect of the extra dimension. The line element becomes

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu + \phi^2 dy^2, \quad (41)$$

and the metric is equivalently

$$g_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & \phi^2 \end{pmatrix}. \quad (42)$$

Before calculating anything, it is useful to understand what ϕ represents. The fifth-dimensional part of the metric is

$$ds_5^2 = \phi^2 dy^2. \quad (43)$$

Suppose that we move only in the fifth direction, such that $dx^\mu = 0$, then the line element becomes simply:

$$ds^2 = \phi^2 dy^2. \quad (44)$$

The physical length around the compact dimension is

$$L = \int_0^{2\pi R} \sqrt{g_{44}} dy, \quad (45)$$

and recalling that $g_{44} = \phi^2$, we get

$$L = \int_0^{2\pi R} \phi dy. \quad (46)$$

Assuming ϕ doesn't depend on y , then

$$L = 2\pi R \phi. \quad (47)$$

Therefore, if ϕ increases, the hidden dimension expands; if ϕ decreases, the hidden dimension contracts. Based on this, the radion is a field that describes fluctuations in the size of the hidden dimensions of the universe [9] [13]. The 5D action is

$$S_5 = \frac{1}{16\pi G_5} \int d^5x \sqrt{-g_5} R_5, \quad (48)$$

and the coordinates are

$$x^A = (x^\mu, y), \quad (49)$$

such that

$$d^5x = d^4x dy. \quad (50)$$

Substituting this into equation (48) yields

$$S_5 = \frac{1}{16\pi G_5} \int d^4x \int dy \sqrt{-g_5} R_5. \quad (51)$$

Now recall that the compact dimension is a circle:

$$0 \leq y < 2\pi R, \quad (52)$$

such that

$$\int dy = 2\pi R. \quad (53)$$

The action becomes:

$$S_5 = \frac{2\pi R}{16\pi G_5} \int d^4x \sqrt{-g_5} R_5 \quad (54)$$

For the constant-radius compactification, it's convenient to define the four-dimensional Newton constant [12] where $\phi \in [0, 2\pi R]$ and R is a fixed geometric constant:

$$G_4 = \frac{G_5}{2\pi R}. \quad (55)$$

Therefore,

$$S_4 = \frac{1}{16\pi G_4} \int d^4x \sqrt{-g_5} R_5. \quad (56)$$

All that remains is to express $\sqrt{-g_5} R_5$ in terms of four-dimensional quantities.

The determinant of a block diagonal satisfies

$$\det \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \det(A) \det(B). \quad (57)$$

Our metric is:

$$g_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & \phi^2 \end{pmatrix}. \quad (58)$$

Therefore,

$$\det(g_5) = \det(g_4) \phi^2. \quad (59)$$

Taking the square root:

$$\sqrt{-g_5} = \sqrt{-g_4} \phi, \quad (60)$$

or

$$\sqrt{-g_5} = \phi \sqrt{-g} \quad (61)$$

The extra dimension contributes a factor of ϕ because the volume of the compact dimension depends on its size. The Ricci scalar is constructed from the curvature tensor [12]:

$$R_5 = g^{AB} R_{AB}. \quad (62)$$

The result of calculating the Christoffel symbols and curvature tensors for the metric

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu + \phi^2 dy^2 \quad (63)$$

is [12]:

$$R_5 = R_4 - 2 \frac{\square \phi}{\phi}, \quad (64)$$

where R_4 is the ordinary four-dimensional Ricci scalar (ignoring electromagnetic forces) when using the signature $(-, +, +, +, +)$, and

$$\square \phi = \nabla^\mu \nabla_\mu \phi \quad (65)$$

is the curved-space wave operator (the D'Alembertian).

Substituting into the action (equation (56)):

$$S_4 = \frac{1}{16\pi G_4} \int d^4x \sqrt{-g} \phi \left(R_4 - 2 \frac{\square \phi}{\phi} \right) \quad (66)$$

and, multiplying out,

$$S_4 = \frac{1}{16\pi G_4} \int d^4x \sqrt{-g} (\phi R_4 - 2 \square \phi). \quad (67)$$

The term

$$\int \sqrt{-g} \square \phi \quad (68)$$

is a boundary term and does not affect the equations of motion through Hamilton's variational principle (assuming appropriate boundary conditions). After applying this, we get

$$S_4 = \frac{1}{16\pi G_4} \int d^4x \sqrt{-g} \phi R_4. \quad (69)$$

The size of the extra dimension has become coupled to ordinary gravity.

The action

$$S = \frac{1}{16\pi G_4} \int d^4x \sqrt{-g} \phi R \quad (70)$$

is not yet in the standard Einstein form because the scalar multiplies the curvature. We redefine the field: let

$$\phi = e^{\sqrt{\frac{2}{3}} \frac{\sigma}{M_{\text{Pl}}}}, \quad (71)$$

where

$$M_{\text{Pl}} = \frac{1}{\sqrt{8\pi G}} \quad (72)$$

is the Planck mass and the field $\sigma(x)$ is the canonically normalised radion.

After transforming to the Einstein frame, the action becomes:

$$S_4 = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} (\partial\sigma)^2 \right] \quad (73)$$

The last few steps are the result of a lengthy derivation, available in [Appendix A](#).

5 The Radion Field as an Effective Cold-Matter Component

In the previous section, we began with a gravitational theory in five dimensions and showed that after compactification, the geometry of the extra dimension produces a scalar field in four dimensions. This scalar field is the radion which represents fluctuations in the size of the hidden dimension. The idea is that although the radion originates from geometry, it mathematically behaves like a physical field. Therefore, it can carry energy, momentum, pressure and contribute to the gravitational field. A four-dimensional observer would not see the extra dimension directly. Instead, they would observe its gravitational influence through the radion's energy density [9]. After dimensional reduction and transforming into the Einstein frame, the radion action takes the form:

$$S_\sigma = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma - V(\sigma) \right). \quad (74)$$

This has the same mathematical form as a standard scalar field in physics [3].

The first term,

$$-\frac{1}{2} g^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma, \quad (75)$$

represents the kinetic energy of the field.

The second term $V(\sigma)$ is the potential energy associated with the configuration of the extra dimension.

Physically if the radius of the extra dimension changes rapidly, the radion has a large kinetic energy, and if the radius is displaced from its preferred size, it stores potential energy. In general relativity, matter and energy are described by the stress-energy tensor $T_{\mu\nu}$. The stress-energy tensor of a field is obtained from its action [3]

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}} \quad (76)$$

For the radion action,

$$S_\sigma = \int d^4x \sqrt{-g} \left(-\frac{1}{2} (\partial\sigma)^2 - V(\sigma) \right). \quad (77)$$

The variation (using product rule on the action) gives

$$T_{\mu\nu}^\sigma = \partial_\mu \sigma \partial_\nu \sigma - g_{\mu\nu} \left(\frac{1}{2} (\partial\sigma)^2 + V(\sigma) \right). \quad (78)$$

This equation describes how the radion contributes to spacetime curvature. To compare the radion with dark matter, we consider a homogeneous universe. The universe is described by the Friedmann–Lemaître–Robertson–Walker (FLRW) metric [8]:

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^2, \quad (79)$$

where $a(t)$ is the scale factor and t is cosmic time.

We assume that the radion is uniform throughout space:

$$\sigma = \sigma(t). \quad (80)$$

Therefore,

$$\partial_i \sigma = 0, \quad (81)$$

and only time evolution remains:

$$\partial_0 \sigma = \dot{\sigma}. \quad (82)$$

The energy density is [15]

$$\rho_\sigma = \frac{1}{2} \dot{\sigma}^2 + V(\sigma), \quad (83)$$

and the pressure is

$$p_\sigma = \frac{1}{2} \dot{\sigma}^2 - V(\sigma). \quad (84)$$

These two standard equations are important because they tell us that the radion can behave differently depending on whether its energy is dominated by kinetic or potential energy. Cosmologists describe different forms of matter using the equation-of-state parameter:

$$w = \frac{p}{\rho} \quad (85)$$

where

- $w \approx 0$ corresponds to matter
- $w = \frac{1}{3}$ corresponds to radiation
- $w = -1$ corresponds to dark energy

For the radion,

$$w_\sigma = \frac{\frac{1}{2} \dot{\sigma}^2 - V(\sigma)}{\frac{1}{2} \dot{\sigma}^2 + V(\sigma)}. \quad (86)$$

Now consider two limiting cases:

If potential energy dominates,

$$V(\sigma) \gg \frac{1}{2} \dot{\sigma}^2, \quad (87)$$

then

$$\rho_\sigma \approx V(\sigma), \quad (88)$$

and

$$p_\sigma \approx -V(\sigma). \quad (89)$$

Therefore,

$$w_\sigma \approx -1. \quad (90)$$

This means the radion behaves like dark energy, causing accelerated expansion rather than acting as dark matter.

If kinetic energy dominates,

$$\frac{1}{2} \dot{\sigma}^2 \gg V(\sigma), \quad (91)$$

then

$$\rho_\sigma \approx \frac{1}{2} \dot{\sigma}^2, \quad (92)$$

and

$$p_\sigma \approx \frac{1}{2}\dot{\sigma}^2, \quad (93)$$

giving

$$w_\sigma \approx 1. \quad (94)$$

This behaves like a stiff fluid, not dark matter. Therefore, neither extreme gives the desired behaviour.

For dark matter, we require

$$w \approx 0, \quad (95)$$

meaning

$$p \approx 0. \quad (96)$$

The evolution of any scalar field in an expanding universe is governed by the Klein-Gordon equation [15] (derived in [Appendix B](#)):

$$\ddot{\sigma} + 3H\dot{\sigma} + \frac{dV}{d\sigma} = 0, \quad (97)$$

where $\ddot{\sigma}$ represents acceleration of the field, $3H\dot{\sigma}$ is the effect of cosmic expansion, and $\frac{dV}{d\sigma}$ is the force from the potential. The term $3H\dot{\sigma}$ acts like friction: as the universe expands, the motion of the radion becomes dampened. The energy density evolves according to:

$$\dot{\rho}_\sigma + 3H(\rho_\sigma + p_\sigma) = 0. \quad (98)$$

This is the cosmological conservation equation [15].

Substituting

$$p = w\rho \quad (99)$$

gives

$$\dot{\rho} + 3H\rho(1 + w) = 0. \quad (100)$$

Solving:

$$\rho \propto a^{-3(1+w)}. \quad (101)$$

This equation allows direct comparison with different cosmic components.

For cold dark matter,

$$w_{DM} = 0. \quad (102)$$

Therefore,

$$\rho_{DM} \propto a^{-3}. \quad (103)$$

The density decreases because the universe expands and matter becomes more spread out [1][8].

For the radion to mimic dark matter, we require:

$$\rho_\sigma \propto a^{-3}, \quad (104)$$

which means

$$w_\sigma \approx 0. \quad (105)$$

The result does not prove that dark matter is caused by extra dimensions. Instead, a universe containing hidden dimensions may produce an effective four-dimensional field whose gravitational influence is almost identical to that of dark matter. From the viewpoint of a four-dimensional observer, hidden dimensions could give rise to the radion field, which could mimic effective gravitational mass.

6 Satisfying Dark Matter Expansion

Let a five-dimensional spacetime admit a compactification of the form

$$M_5 = M_4 \times S^1, \quad (106)$$

and suppose that dimensional reduction produces a canonically normalised radion field, σ , with effective potential $V(\sigma)$.

Assume:

- The compactification is independent of the extra coordinate,

$$\partial_y g_{AB} = 0. \quad (107)$$

- The radion potential possesses a stable minimum at $\sigma = \sigma_0$,

$$V'(\sigma_0) = 0, \quad V''(\sigma_0) > 0 \quad (108)$$

- The radion mass near the minimum is defined by $m_\sigma^2 = V''(\sigma_0)$ during the epoch of interest, $m_\sigma \gg H$.
- The oscillation amplitude is sufficiently small that the potential is well approximated by its quadratic expansion about $\sigma = \sigma_0$.

If all of these assumptions are made, then the oscillatory component of the radion has the averaged equation of state:

$$\langle w_\sigma \rangle \simeq 0, \quad (109)$$

and its averaged energy density scales as:

$$\langle \rho_\sigma \rangle \propto a^{-3}. \quad (110)$$

Consequently, the oscillating radion is dynamically the same as pressureless matter at the level of the homogeneous cosmological background.

The effective radion action is:

$$S_\sigma = \int d^4x \sqrt{-g} \left[-\frac{1}{2}(\partial\sigma)^2 - V(\sigma) \right]. \quad (111)$$

The corresponding energy density and pressure are:

$$\rho_\sigma = \frac{1}{2}\dot{\sigma}^2 + V(\sigma), \quad (112)$$

$$p_\sigma = \frac{1}{2}\dot{\sigma}^2 - V(\sigma). \quad (113)$$

Expanding around the stable minimum,

$$\sigma = \sigma_0 + \delta\sigma, \quad (114)$$

and using

$$V'(\sigma_0) = 0, \quad (115)$$

the potential has local expansion (from Taylor's Theorem):

$$V(\sigma_0) + \frac{1}{2}m_\sigma^2(\delta\sigma)^2 + O(\delta\sigma^3). \quad (116)$$

The constant term $V(\sigma_0)$ behaves as a vacuum-energy contribution. Assuming the oscillation amplitude is sufficiently small that the higher-order terms can be neglected, the oscillatory component obeys

$$\ddot{\delta\sigma} + 3H\dot{\delta\sigma} + m_\sigma^2\delta\sigma = 0. \quad (117)$$

When

$$m_\sigma \gg H, \quad (118)$$

the field undergoes many oscillations during a Hubble time [16][17]. The expansion is therefore slowed on the timescale of an individual oscillation, and the field may be written approximately as:

$$\delta\sigma = A(t) \cos(m_\sigma t), \quad (119)$$

where the amplitude $A(t)$ varies slowly compared to the oscillation period. Averaging over one oscillation gives:

$$\langle \dot{\delta\sigma}^2 \rangle \simeq m_\sigma^2 \langle (\delta\sigma)^2 \rangle. \quad (120)$$

Hence,

$$\left\langle \frac{1}{2} \dot{\delta\sigma}^2 \right\rangle \simeq \left\langle \frac{1}{2} m_\sigma^2 (\delta\sigma)^2 \right\rangle. \quad (121)$$

The averaged pressure of the oscillating component is therefore:

$$\langle p_{\text{osc}} \rangle = \left\langle \frac{1}{2} \dot{\delta\sigma}^2 \right\rangle - \left\langle \frac{1}{2} m_\sigma^2 (\delta\sigma)^2 \right\rangle \simeq 0. \quad (122)$$

Thus,

$$\langle w_{\text{osc}} \rangle \simeq 0. \quad (123)$$

The continuity equation,

$$\dot{\rho}_{\text{osc}} + 3H(\rho_{\text{osc}} + p_{\text{osc}}) = 0, \quad (124)$$

then becomes:

$$\langle \dot{\rho}_{\text{osc}} \rangle + 3H\rho_{\text{osc}} = 0. \quad (125)$$

Since

$$H = \frac{\dot{a}}{a}, \quad (126)$$

the continuity equation yields:

$$\rho_{\text{osc}} \propto a^{-3}. \quad (127)$$

7 Radion Potential Mechanics & Harmonic Oscillations

For simplicity, we consider a universe containing only ordinary matter and the radion field to determine whether the radion can exhibit the characteristic background evolution of cold dark matter. Radiation, baryonic physics, and other cosmological components are neglected because they are not essential for the purpose of this toy model. The aim is not to reproduce the complete expansion history of the Universe, but to determine whether the radion can exhibit the characteristic background evolution of cold dark matter. In the previous sections, we showed that Kaluza–Klein compactification produces a scalar field, the radion $\sigma(x)$, which represents changes in the size of an extra spatial dimension.

The radion contributes to the Einstein equations through its stress-energy tensor:

$$G_{\mu\nu} = 8\pi G(T_{\mu\nu}^{\text{visible}} + T_{\mu\nu}^{\sigma}) \quad (128)$$

The dynamics of the radion depends on its potential, $V(\sigma)$, which describes the energy associated with different sizes of the extra dimension.

A realistic compactification mechanism would need a potential with a stable minimum. This essentially means that the extra dimension should prefer a particular size.

The simplest possible stable potential is a quadratic potential (derived in [Appendix B](#)):

$$V(\sigma) = \frac{1}{2}m_{\sigma}^2\sigma^2 \quad (129)$$

where $\sigma = 0$ corresponds to the preferred size of the extra dimension, and the value m_{σ} is the value of the mass of the radion particle. However, this mass changes depending on which mechanism you use for interactions that generate mass. We assumed the simplest version of Kaluza–Klein, where our radion is massless and where the size of the extra dimension is then a degree of freedom. $V(\sigma)$ arises from an unspecified stabilization mechanism (e.g., Goldberger–Wise mechanism or flux compactification) detailed in [Appendix B](#).

If the radion is displaced from equilibrium,

$$\sigma \neq 0, \quad (130)$$

the field experiences a restoring force:

$$F = -\frac{dV}{d\sigma}, \quad (131)$$

which gives

$$F = -m_{\sigma}^2\sigma. \quad (132)$$

Notice how this is similar to simple harmonic motion?

We model the universe using the Friedmann–Lemaître–Robertson–Walker metric:

$$ds^2 = -dt^2 + a(t)^2 dx^2. \quad (133)$$

The expansion of the universe is governed by the Friedmann equation:

$$H^2 = \frac{8\pi G}{3}(\rho_m + \rho_{\sigma}), \quad (134)$$

where:

- $H = \frac{\dot{a}}{a}$ is the Hubble parameter
- ρ_m is ordinary matter density
- ρ_σ is radion density

The radion evolves according to the Klein–Gordon equation:

$$\ddot{\sigma} + 3H\dot{\sigma} + \frac{dV}{d\sigma} = 0. \quad (135)$$

Substituting

$$V(\sigma) = \frac{1}{2}m_\sigma^2\sigma^2 \quad (136)$$

gives:

$$\ddot{\sigma} + 3H\dot{\sigma} + m_\sigma^2\sigma = 0. \quad (137)$$

The expansion of the universe acts like friction on the oscillating extra dimension.

At very early times,

$$H \gg m_\sigma, \quad (138)$$

meaning that cosmic expansion is faster than radion oscillations. The friction term dominates,

$$3H\dot{\sigma} \gg m_\sigma^2\sigma, \quad (139)$$

so the radion is approximately frozen:

$$\dot{\sigma} \approx 0, \quad (140)$$

and its energy is dominated by its potential:

$$\rho_\sigma \approx V(\sigma), \quad (141)$$

giving

$$w \approx -1. \quad (142)$$

Therefore, at very early times, the radion behaves temporarily like vacuum energy.

However, as the universe expands,

$$H \rightarrow 0, \quad (143)$$

and eventually,

$$m_\sigma > H, \quad (144)$$

the radion begins oscillating. This is only reasonable for the homogeneous field, but there is an important qualification: the field only behaves like dark matter after it begins oscillating.

When

$$m_\sigma \gg H. \quad (145)$$

the expansion of the universe is slow compared with the oscillation.

Ignoring the friction term momentarily,

$$\ddot{\sigma} + m_\sigma^2\sigma = 0, \quad (146)$$

admits the solution:

$$\sigma(t) = \sigma_0 \cos(m_\sigma t). \quad (147)$$

The radion oscillates around the minimum

$$\sigma = 0. \quad (148)$$

The energy density is

$$\rho_\sigma = \frac{1}{2}\dot{\sigma}^2 + \frac{1}{2}m_\sigma^2\sigma^2. \quad (149)$$

For

$$\sigma(t) = \sigma_0 \cos(m_\sigma t), \quad (150)$$

the kinetic term is

$$\dot{\sigma} = -m_\sigma\sigma_0 \sin(m_\sigma t). \quad (151)$$

Therefore:

$$\rho_\sigma = \frac{1}{2}m_\sigma^2\sigma_0^2 \sin^2(m_\sigma t) + \frac{1}{2}m_\sigma^2\sigma_0^2 \cos^2(m_\sigma t) \quad (152)$$

since

$$\sin^2 x + \cos^2 x = 1, \quad (153)$$

and we obtain:

$$\rho_\sigma = \frac{1}{2}m_\sigma^2\sigma_0^2. \quad (154)$$

The total energy remains constant without expansion.

However, in an expanding universe, the oscillation amplitude decreases [16][17]. The averaged equation of state becomes:

$$\langle w \rangle = 0, \quad (155)$$

which means

$$\langle \rho_\sigma \rangle \propto a^{-3}. \quad (156)$$

This is exactly the same scaling as cold dark matter.

Since time-averaged pressure vanishes over many oscillation periods,

$$\langle p \rangle = \langle \frac{1}{2}\dot{\sigma}^2 \rangle - \langle V(\sigma) \rangle \approx 0, \quad (157)$$

we find that the average kinetic energy equals the average potential energy:

$$\langle \frac{1}{2}\dot{\sigma}^2 \rangle \approx \langle V(\sigma) \rangle. \quad (158)$$

While instantaneous energy continuously changes between kinetic and potential forms throughout each cycle, their equality on average gives rise to the effective pressureless matter behaviour, $\langle w \rangle \approx 0$. To compare with cosmology, we solve the coupled equations:

$$\ddot{\sigma} + 3H\dot{\sigma} + m_\sigma^2\sigma = 0, \quad (159)$$

and

$$H^2 = \frac{8\pi G}{3}[\rho_m + \frac{1}{2}\dot{\sigma}^2 + \frac{1}{2}m_\sigma^2\sigma^2]. \quad (160)$$

A convenient dimensionless form uses

$$N = \ln(a) \quad (161)$$

as the time variable.

The equations become:

$$\frac{d\sigma}{dN} = \frac{\dot{\sigma}}{H}, \quad (162)$$

and

$$\frac{d\dot{\sigma}}{dN} = -3\dot{\sigma} - \frac{m_\sigma^2 \sigma}{H}. \quad (163)$$

These equations can be solved numerically for different radion masses. The quantity we compare is:

$$\frac{\rho_\sigma(a)}{\rho_{\sigma,0}} \quad (164)$$

against

$$a^{-3}. \quad (165)$$

The qualitative evolution is $H \gg m_\sigma$ for the early universe. The radion is frozen such that $\rho_\sigma \approx \text{constant}$ and behaves like dark energy. Then, there is an intermediate period where $H \approx m_\sigma$, and the radion begins oscillating. The late universe behaviour is $H < m_\sigma$, where the oscillating radion behaves as $\rho_\sigma \propto a^{-3}$, matching cold dark matter [16][17].

The standard cosmological model requires approximately $\Omega_{\text{DM}} \approx 0.265$ [1]. The toy model must satisfy, with suitable initial conditions:

$$\rho_\sigma(a) \approx \rho_{\text{DM}}(a) \quad (166)$$

over cosmological timescales. Therefore, a four-dimensional observer could interpret the radion as dark matter given that it satisfies this condition.

8 Discussion and Conclusion

The aim of this paper was not to prove that dark matter is definitely caused by hidden dimensions, but to investigate a more precise question: can the geometry of extra dimensions produce a gravitational effect similar to dark matter?

Starting from five-dimensional general relativity, we found that compactifying an extra spatial dimension produces a scalar field in four dimensions: the radion.

The radion contributes to Einstein's equations as

$$G_{\mu\nu} = 8\pi G (T_{\mu\nu}^{\text{visible}} + T_{\mu\nu}^\sigma), \quad (167)$$

meaning that although the source is geometric rather than particle-based, its gravitational effect can be identical to that of matter.

The key test was whether

$$\rho_\sigma \propto a^{-3}, \quad (168)$$

because this is characteristic background behaviour of cold dark matter [8].

For a quadratic radion potential:

$$V(\sigma) = \frac{1}{2}m_\sigma^2\sigma^2, \quad (169)$$

we found that once the radion begins oscillating,

$$m_\sigma > H, \quad (170)$$

its average equation of state becomes:

$$\langle w \rangle \approx 0, \quad (171)$$

giving

$$\langle \rho_\sigma \rangle \propto a^{-3}. \quad (172)$$

This is the same scaling as cold dark matter. Therefore, the answer to our research inquiry is yes, Kaluza–Klein geometry can mimic dark matter at the level of cosmic expansion.

A four-dimensional observer would see an additional gravitational contribution behaving almost exactly like invisible matter, even though its origin is the dynamics of a hidden dimension. The conventional dark matter model assumes that the universe contains a new particle species [18].

One of the strongest aspects of the model is that the radion is not artificially introduced; it appears automatically from the five-dimensional metric:

$$g_{44} = \phi^2. \quad (173)$$

It also adheres to the Einsteinian heuristic of simplicity: the solution to dark matter is in geometry, not through adding a new particle. Kaluza–Klein theory is historically important because it demonstrates that different forces can emerge from geometry.

The same five-dimensional metric produces:

$$\text{gravity} + \text{electromagnetism} + \text{scalar fields} \quad (174)$$

This idea later influenced string theory, where additional dimensions are required for consistency [19].

Therefore, studying Kaluza–Klein dark matter is not an isolated idea; it connects to a broader attempt to unify fundamental physics.

A useful scientific model should be testable. The radion predicts:

- a scalar field
- a specific energy evolution
- possible interactions with ordinary matter

Future experiments could search for deviations from general relativity caused by radion interactions.

Although the toy model gives an interesting result, it is not yet a complete dark matter theory. Several major challenges remain.

The first challenge is: why does the extra dimension remain the same size today?

The radion describes fluctuations:

$$R \rightarrow R + \delta R, \quad (175)$$

but observations suggest that any change in fundamental constants must be extremely small. A realistic model, therefore, requires a stabilising mechanism that creates a potential $V(\sigma)$ with a stable minimum [20]. Without this, the hidden dimension could continuously expand or shrink.

The model successfully reproduces

$$\rho \propto a^{-3}, \quad (176)$$

which describes cosmic evolution. However, dark matter is also required to explain [1]:

- galaxy rotation curves
- gravitational lensing
- galaxy cluster dynamics
- structure formation

A complete Kaluza–Klein model must show that the radion forms the correct gravitational structures on smaller scales.

Further, a scalar field coupled too strongly to ordinary matter would produce detectable fifth forces. The radion must interact weakly enough that Newtonian gravity remains accurate, planetary motion remains unchanged, and laboratory tests of gravity are satisfied. This requires careful model-building.

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Appendix A: Einstein Frame Derivation

From the previous section, after reducing the five-dimensional Einstein–Hilbert action with

$$ds^2 = g_{\mu\nu}^{(J)} dx^\mu dx^\nu + \phi^2 dy^2 \quad (\text{A1})$$

and dropping the boundary term, we obtained [21]

$$S_4 = \frac{1}{16\pi G_4} \int d^4x \sqrt{-g_J} \phi R_J. \quad (\text{A2})$$

Here I have deliberately written J on the metric:

$$g_{\mu\nu}^{(J)} \quad (\text{A3})$$

This is the Jordan-frame metric.

We define the reduced Planck mass by

$$M_{\text{Pl}}^2 = \frac{1}{8\pi G_4}. \quad (\text{A4})$$

Therefore,

$$\frac{1}{16\pi G_4} = \frac{M_{\text{Pl}}^2}{2}. \quad (\text{A5})$$

Thus, the action becomes:

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_J} \phi R_J. \quad (\text{A6})$$

This is our starting point.

The standard Einstein–Hilbert action in four dimensions is:

$$S_{\text{EH}} = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g} R. \quad (\text{A7})$$

Notice that the Ricci scalar appears by itself.

But our action is:

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_J} \phi R_J. \quad (\text{A8})$$

The factor ϕ multiplies the curvature. Therefore, the gravitational coupling effectively depends on the radion:

$$G_{\text{eff}} \sim \frac{G_4}{\phi}. \quad (\text{A9})$$

We want to perform a conformal transformation that removes this factor. We define a new metric $g_{\mu\nu}^{(E)}$, called the Einstein-frame metric, by

$$g_{\mu\nu}^{(E)} = \phi g_{\mu\nu}^{(J)} \quad (\text{A10})$$

or equivalently [21],

$$g_{\mu\nu}^{(J)} = \phi^{-1} g_{\mu\nu}^{(E)}. \quad (\text{A11})$$

The superscript E means Einstein frame. This transformation is sometimes called a Weyl transformation or conformal transformation. The important point is that we are not changing the physical number of degrees of freedom. We are simply changing the four-dimensional metric variable used to describe them.

Since

$$g_{\mu\nu}^{(J)} = \phi^{-1} g_{\mu\nu}^{(E)}, \quad (\text{A12})$$

the inverse metric satisfies

$$g_J^{\mu\nu} = \phi g_E^{\mu\nu}. \quad (\text{A13})$$

Therefore,

$$g_J^{\mu\nu} = \phi g_E^{\mu\nu}. \quad (\text{A14})$$

This will be useful when dealing with derivatives of ϕ . In four dimensions, if

$$g_{\mu\nu}^{(J)} = \phi^{-1} g_{\mu\nu}^{(E)}, \quad (\text{A15})$$

then every component of the metric has been multiplied by ϕ^{-1} . For a 4×4 metric, the determinant therefore transforms as:

$$g_J = \phi^{-4} g_E. \quad (\text{A16})$$

Taking the square root gives:

$$\sqrt{-g_J} = \phi^{-2} \sqrt{-g_E}. \quad (\text{A17})$$

Thus,

$$\sqrt{-g_J} = \phi^{-2} \sqrt{-g_E}. \quad (\text{A18})$$

This is one of the key steps. Next, we need the transformation of the Ricci scalar. For a conformal transformation in four dimensions,

$$g_{\mu\nu}^{(E)} = \phi g_{\mu\nu}^{(J)}, \quad (\text{A19})$$

the Ricci scalar transforms (after expanding and simplifying the standard conformal transformation of the Ricci scalar in four dimensions [22]) according to:

$$R_J = \phi \left[R_E + 3\Box_E \ln \phi - \frac{3}{2}(\partial \ln \phi)^2 E^2 \right]. \quad (\text{A20})$$

Here,

$$\Box_E \ln \phi = g_E^{\mu\nu} \nabla_\mu^{(E)} \nabla_\nu^{(E)} \ln \phi, \quad (\text{A21})$$

and

$$(\partial \ln \phi)^2 E^2 = g_E^{\mu\nu} \partial_\mu \ln \phi \partial_\nu \ln \phi. \quad (\text{A22})$$

So, we have

$$R_J = \phi \left[R_E + 3\Box_E \ln \phi - \frac{3}{2}(\partial \ln \phi)^2 E^2 \right]. \quad (\text{A23})$$

This is the crucial geometric identity. We started with:

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_J} \phi R_J. \quad (\text{A24})$$

We have

$$\sqrt{-g_J} = \phi^{-2} \sqrt{-g_E} \quad (\text{A25})$$

and

$$R_J = \phi \left[R_E + 3\Box_E \ln \phi - \frac{3}{2}(\partial \ln \phi)^2 \right]. \quad (\text{A26})$$

Therefore,

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x [\phi^{-2} \sqrt{-g_E}] \phi \left[\phi \left(R_E + 3\Box_E \ln \phi - \frac{3}{2}(\partial \ln \phi)^2 \right) \right]. \quad (\text{A27})$$

Now, look at the powers of ϕ :

$$\phi^{-2} \times \phi \times \phi = 1. \quad (\text{A28})$$

Therefore, all factors of ϕ multiplying the Ricci scalar disappear. We obtain:

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_E} \left[R_E + 3\Box_E \ln \phi - \frac{3}{2}(\partial \ln \phi)^2 \right]. \quad (\text{A29})$$

So,

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_E} \left[R_E + 3\Box_E \ln \phi - \frac{3}{2}(\partial \ln \phi)^2 \right]. \quad (\text{A30})$$

Consider:

$$\int d^4x \sqrt{-g_E} \Box_E \ln \phi. \quad (\text{A31})$$

Since [23]

$$\Box_E \ln \phi = \nabla_\mu \nabla^\mu \ln \phi, \quad (\text{A32})$$

we can write [3, 18]

$$\sqrt{-g_E} \Box_E \ln \phi = \partial_\mu (\sqrt{-g_E} \nabla^\mu \ln \phi). \quad (\text{A33})$$

Therefore,

$$\int d^4x \sqrt{-g_E} \Box_E \ln \phi \quad (\text{A34})$$

is a total derivative. Assuming appropriate boundary conditions, it does not contribute to the bulk equations of motion. Thus we can drop it via Hamilton's Variational Principle [24]. The action becomes:

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_E} \left[R_E - \frac{3}{2}(\partial \ln \phi)^2 \right]. \quad (\text{A35})$$

So, the Einstein-frame action is:

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_E} \left[R_E - \frac{3}{2} g_E^{\mu\nu} \partial_\mu \ln \phi \partial_\nu \ln \phi \right] \quad (\text{A36})$$

This is the key intermediate result.

Originally, the radion appeared as:

$$\phi R_J. \quad (\text{A37})$$

It did not initially appear as an ordinary scalar kinetic term. After the Weyl transformation, we have

$$-\frac{3}{4}M_{\text{Pl}}^2(\partial \ln \phi)^2. \quad (\text{A38})$$

Therefore, the dynamics of the compactification radius have become the dynamics of a four-dimensional scalar field. We can write the action as:

$$S_4 = \int d^4x \sqrt{-g_E} \left[\frac{M_{\text{Pl}}^2}{2} R_E - \frac{3M_{\text{Pl}}^2}{4} (\partial \ln \phi)^2 \right]. \quad (\text{A39})$$

Now, we simply need to choose a field variable whose kinetic term has the canonical form

$$-\frac{1}{2}(\partial \sigma)^2. \quad (\text{A40})$$

We want

$$-\frac{3M_{\text{Pl}}^2}{4}(\partial \ln \phi)^2 \quad \text{to equal} \quad -\frac{1}{2}(\partial \sigma)^2. \quad (\text{A41})$$

Therefore, we require:

$$\frac{1}{2}(\partial \sigma)^2 = \frac{3M_{\text{Pl}}^2}{4}(\partial \ln \phi)^2. \quad (\text{A42})$$

Multiply both sides by 2:

$$(\partial \sigma)^2 = \frac{3}{2}M_{\text{Pl}}^2(\partial \ln \phi)^2. \quad (\text{A43})$$

Therefore,

$$\partial_\mu \sigma = \sqrt{\frac{3}{2}}M_{\text{Pl}} \partial_\mu \ln \phi, \quad (\text{A44})$$

and integrating:

$$\sigma = \sqrt{\frac{3}{2}}M_{\text{Pl}} \ln \phi + \text{constant}. \quad (\text{A45})$$

We can choose the integration constant to be zero, giving:

$$\sigma = \sqrt{\frac{3}{2}}M_{\text{Pl}} \ln \phi. \quad (\text{A46})$$

This is the canonically normalised radion field.

Starting from

$$\sigma = \sqrt{\frac{3}{2}}M_{\text{Pl}} \ln \phi, \quad (\text{A47})$$

divide by M_{Pl} :

$$\frac{\sigma}{M_{\text{Pl}}} = \sqrt{\frac{3}{2}} \ln \phi. \quad (\text{A48})$$

Therefore,

$$\ln \phi = \sqrt{\frac{2}{3}} \frac{\sigma}{M_{\text{Pl}}}. \quad (\text{A49})$$

Exponentiating,

$$\phi = \exp \left(\sqrt{\frac{2}{3}} \frac{\sigma}{M_{\text{Pl}}} \right). \quad (\text{A50})$$

This is exactly the exponential relation we wrote, but now we have derived it. We have:

$$\partial_\mu \ln \phi = \sqrt{\frac{2}{3}} \frac{1}{M_{\text{Pl}}} \partial_\mu \sigma. \quad (\text{A51})$$

Therefore,

$$(\partial \ln \phi)^2 = \frac{2}{3M_{\text{Pl}}^2} (\partial \sigma)^2. \quad (\text{A52})$$

Substituting this into

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_E} \left[R_E - \frac{3}{2} (\partial \ln \phi)^2 \right], \quad (\text{A53})$$

we obtain

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_E} \left[R_E - \frac{3}{2} \left(\frac{2}{3M_{\text{Pl}}^2} (\partial \sigma)^2 \right) \right]. \quad (\text{A54})$$

The numerical factors simplify:

$$\frac{3}{2} \times \frac{2}{3} = 1. \quad (\text{A55})$$

Therefore,

$$S_4 = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_E} \left[R_E - \frac{1}{M_{\text{Pl}}^2} (\partial \sigma)^2 \right]. \quad (\text{A56})$$

Distributing the factor,

$$S_4 = \int d^4x \sqrt{-g_E} \left[\frac{M_{\text{Pl}}^2}{2} R_E - \frac{1}{2} (\partial \sigma)^2 \right]. \quad (\text{A57})$$

Thus,

$$S_4 = \int d^4x \sqrt{-g_E} \left[\frac{M_{\text{Pl}}^2}{2} R_E - \frac{1}{2} g_E^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma \right]. \quad (\text{A58})$$

And this is precisely the canonical Einstein-scalar action.

The original radion ϕ appeared geometrically as:

$$ds_5^2 = g_{\mu\nu}^{(J)} dx^\mu dx^\nu + \phi^2 dy^2. \quad (\text{A59})$$

Recall that the physical circumference of the compact dimension is

$$L = 2\pi R\phi. \quad (\text{A60})$$

Therefore, $\phi(x)$ describes fluctuations in the physical size of the fifth dimension. We then introduced

$$\sigma(x) = \sqrt{\frac{3}{2}} M_{\text{Pl}} \ln \phi(x). \quad (\text{A61})$$

So, σ is simply a convenient field variable describing those geometrical fluctuations, chosen so that its kinetic energy has the standard canonical form, with:

$$\phi = e^{\sqrt{2/3} \sigma / M_{\text{Pl}}} \quad (\text{A62})$$

Appendix B: Potential Derivation in the Einstein Frame

The potential

$$V(\sigma) = \frac{1}{2}m_\sigma^2\sigma^2 \quad (\text{B1})$$

was introduced without an explanation of where it comes from. There are two stages: Kaluza–Klein reduction gives a massless radion, then additional compactification/stabilisation gives the radion a potential and, therefore, a mass.

We started with:

$$S_5 = \frac{1}{16\pi G_5} \int d^5x \sqrt{-g_5} R_5. \quad (\text{B2})$$

and the metric

$$ds_5^2 = g_{\mu\nu} dx^\mu dx^\nu + \phi^2 dy^2. \quad (\text{B3})$$

After dimensional reduction and transformation to the Einstein frame, we obtained

$$S_4 = \int d^4x \sqrt{-g_E} \left[\frac{M_{\text{Pl}}^2}{2} R_E - \frac{1}{2} (\partial\sigma)^2 \right]. \quad (\text{B4})$$

This is the action of a massless scalar field [9].

We can make this particularly clear by comparing it with the general scalar action:

$$S_\sigma = \int d^4x \sqrt{-g_E} \left[-\frac{1}{2} (\partial\sigma)^2 - V(\sigma) \right]. \quad (\text{B5})$$

Our dimensional reduction has produced $V(\sigma) = 0$. Under the assumptions we have made, we set aside the Einstein-Hilbert term and just analyse the action on the scalar field to obtain the potential.

Therefore, the pure theory gives $m_\sigma = 0$ as well. The reason is geometrical. We assumed the fifth dimension is a circle: S^1 . The circle can have an arbitrary radii and, in the simple vacuum theory we've considered, there is no term in the action that tells the radius which value it “prefers”, and the radius is a degree of freedom. The radion describes motion along this parameter. Schematically, different ϕ bring about different sizes of S^1 . If all these configurations have the same energy, then there is no restoring force. Without a restoring force,

$$V'(\sigma) = 0 \quad (\text{B6})$$

$$V(\sigma) = \text{constant} \quad (\text{B7})$$

The constant can be absorbed into a cosmological constant, so in the simplest case we take $V(\sigma) = 0$.

A massless homogeneous scalar does not automatically behave like cold dark matter. For a scalar,

$$\rho_\sigma = \frac{1}{2}\dot{\sigma}^2 + V(\sigma) \quad (\text{B8})$$

and

$$p_\sigma = \frac{1}{2}\dot{\sigma}^2 - V(\sigma). \quad (\text{B9})$$

If $V(\sigma) = 0$, then

$$\rho_\sigma = \frac{1}{2}\dot{\sigma}^2, \quad \text{and} \quad p_\sigma = \frac{1}{2}\dot{\sigma}^2, \quad (\text{B10})$$

and $w_\sigma = 1$. This is a stiff fluid, not cold dark matter, so the pure massless radion does not solve our problem. This is a useful result because it prevents us from making an unjustified claim. To obtain dark-matter-like behaviour, it is required of the extra dimension that it have a preferred size. Suppose the compactification dynamics generate an effective potential $V(\phi)$, then there is some preferred value, $\phi = \phi_0$, at which

$$\left. \frac{dV}{d\phi} \right|_{\phi_0} = 0, \quad (\text{B11})$$

If this point is stable, then

$$\left. \frac{d^2V}{d\phi^2} \right|_{\phi_0} > 0, \quad (\text{B12})$$

The physical meaning is: if the extra dimension becomes slightly larger or smaller than its equilibrium size, a restoring force drives it back toward equilibrium. That restoring force is exactly what allows the radion to oscillate.

We already derived

$$\phi = e^{\sqrt{2/3}\sigma/M_{\text{Pl}}}. \quad (\text{B13})$$

Therefore, if some higher-dimensional physics produces $V(\phi)$, then in terms of the canonical radion,

$$V(\sigma) = V\left(e^{\sqrt{2/3}\sigma/M_{\text{Pl}}}\right) \quad (\text{B14})$$

up to the appropriate Einstein-frame transformation of the potential.

There is an important detail here: if the potential is originally written in the Jordan frame, it does not simply remain $V(\phi)$ after the Weyl transformation. Because $\sqrt{-g_J} = \phi^{-2}\sqrt{-g_E}$, a Jordan-frame potential term,

$$S_V^{(J)} = - \int d^4x \sqrt{-g_J} U(\phi), \quad (\text{B15})$$

becomes

$$S_V^{(J)} = - \int d^4x \sqrt{-g_E} \phi^{-2} U(\phi). \quad (\text{B16})$$

Therefore, the Einstein-frame potential is:

$$V_E(\phi) = \phi^{-2} U(\phi), \quad (\text{B17})$$

and consequently:

$$V_E(\sigma) = e^{-2\sqrt{2/3}\sigma/M_{\text{Pl}}} U\left(e^{\sqrt{2/3}\sigma/M_{\text{Pl}}}\right). \quad (\text{B18})$$

This would be important if we were to later construct a specific stabilisation mechanism.

Suppose the Einstein-frame potential has a stable minimum at $\sigma = \sigma_0$. Then $V'(\sigma_0) = 0$ and $V''(\sigma_0) > 0$.

We can Taylor expand the potential around the minimum:

$$V(\sigma) = V(\sigma_0) + V'(\sigma_0)(\sigma - \sigma_0) + \frac{1}{2}V''(\sigma_0)(\sigma - \sigma_0)^2 + \dots \quad (\text{B19})$$

Because σ_0 is a minimum, $V'(\sigma_0) = 0$.

Therefore,

$$V(\sigma) \simeq V(\sigma_0) + \frac{1}{2}V''(\sigma_0)(\sigma - \sigma_0)^2. \quad (\text{B20})$$

Define the displacement $\delta\sigma = \sigma - \sigma_0$. Then,

$$V(\sigma) \simeq V(\sigma_0) + \frac{1}{2}V''(\sigma_0)(\delta\sigma)^2. \quad (\text{B21})$$

Define

$$m_\sigma^2 = V''(\sigma_0). \quad (\text{B22})$$

Giving us,

$$V(\sigma) \simeq V_0 + \frac{1}{2}m_\sigma^2(\delta\sigma)^2. \quad (\text{B23})$$

If we choose the vacuum energy at the minimum to be zero ($V_0 = 0$), then

$$V(\sigma) \simeq \frac{1}{2}m_\sigma^2(\delta\sigma)^2. \quad (\text{B24})$$

For simplicity, we can redefine the field so that the minimum occurs at $\sigma_0 = 0$. Then,

$$V(\sigma) \simeq \frac{1}{2}m_\sigma^2\sigma^2. \quad (\text{B25})$$

This is where the quadratic potential comes from. This is an important conceptual point. We should not say: "Kaluza–Klein reduction predicts a radion mass m_σ ". Pure K-K reduction does not do that under our assumptions.

Instead:

$$m_\sigma^2 = V''(\sigma_0) \quad (\text{B26})$$

is determined by whatever physical mechanism stabilises the compactification.

So, the logic is:

$$\begin{aligned} \text{KK reduction} &\Rightarrow \text{radion exists} \\ \text{stabilisation physics} &\Rightarrow m_\sigma \neq 0 \end{aligned}$$

Then $m_\sigma \neq 0$ allows the radion to oscillate. Once stabilisation has generated a potential, the Einstein-frame action is:

$$S_4 = \int d^4x \sqrt{-g_E} \left[\frac{M_{\text{Pl}}^2}{2} R_E - \frac{1}{2} g_E^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma - V(\sigma) \right]. \quad (\text{B27})$$

Near the stable minimum, $V(\sigma) \simeq \frac{1}{2}m_\sigma^2\sigma^2$. Therefore,

$$S_4 \simeq \int d^4x \sqrt{-g_E} \left[\frac{M_{\text{Pl}}^2}{2} R_E - \frac{1}{2} (\partial\sigma)^2 - \frac{1}{2} m_\sigma^2 \sigma^2 \right]. \quad (\text{B28})$$

Now we have a standard massive scalar field.

The scalar-field part of the action is:

$$S_\sigma = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma - V(\sigma) \right]. \quad (\text{B29})$$

Varying with respect to σ gives

$$\square\sigma - V'(\sigma) = 0, \quad (\text{B30})$$

with the sign convention depending on the metric signature. For the mostly-plus convention $(-, +, +, +)$, this becomes the cosmological Klein–Gordon equation:

$$\ddot{\sigma} + 3H\dot{\sigma} + V'(\sigma) = 0. \quad (\text{B31})$$

For $V(\sigma) = \frac{1}{2}m_\sigma^2\sigma^2$, we have $V'(\sigma) = m_\sigma^2\sigma$. Therefore,

$$\ddot{\sigma} + 3H\dot{\sigma} + m_\sigma^2\sigma = 0. \quad (\text{B32})$$

This is precisely the equation we need for the dark-matter analysis. Ignore cosmic expansion momentarily ($H = 0$), such that

$$\ddot{\sigma} + m_\sigma^2\sigma = 0. \quad (\text{B33})$$

This is simply the harmonic oscillator equation. Its solution is:

$$\sigma(t) = \sigma_i \cos(m_\sigma t + \delta). \quad (\text{B34})$$

The radion, therefore, oscillates around the stable compactification radius. With cosmic expansion included,

$$\ddot{\sigma} + 3H\dot{\sigma} + m_\sigma^2\sigma = 0. \quad (\text{B35})$$

The term $3H\dot{\sigma}$ acts as Hubble friction. When $m_\sigma \gg H$, the oscillations are much faster than the expansion. In that regime, the approximate solution is:

$$\sigma(t) \simeq A(t) \cos(m_\sigma t + \delta) \quad (\text{B36})$$

with a slowly varying amplitude.

One finds:

$$A(t) \propto a^{-3/2}. \quad (\text{B37})$$

Appendix C: Corollary: The Role of the Quadratic Minimum

The matter-like behaviour is a special property of an approximately quadratic potential.

More generally, for coherent oscillations in a potential of the form

$$V(\sigma) \propto |\sigma|^n, \quad (\text{C1})$$

the averaged equation of state is [16][17]

$$\langle w \rangle = \frac{n-2}{n+2}. \quad (\text{C2})$$

So,

$$\rho_\sigma \propto a^{-6n/(n+2)}. \quad (\text{C3})$$

For the quadratic case ($n=2$),

$$\langle w \rangle = 0, \quad \rho_\sigma \propto a^{-3} \quad (\text{C4})$$

which is precisely the scaling required for cold dark matter.

This shows why an approximately quadratic region of the stabilised radion potential is the relevant condition for matter-like behaviour. The proposition does not establish that every radion is a dark-matter candidate. Rather, it identifies a specific dynamical regime in which a radion can reproduce the background expansion associated with pressureless matter.

Black-Hole No-Hair, Hawking Radiation, and Unitarity: Investigating the Information Paradox

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Abstract

Black holes bring together general relativity, thermodynamics, and quantum mechanics in a way that exposes one of the deepest unresolved problems in modern physics: the black hole information paradox. Classically, the no-hair principle states that a stationary black hole is completely described by its mass, angular momentum, and electric charge. This description leaves no externally visible record of the detailed matter that formed it. Quantum mechanics, however, requires the evolution of a closed system to remain unitary, so information contained in an initial quantum state cannot be fundamentally destroyed. Hawking radiation creates a tension between these ideas because black holes are predicted to evaporate through approximately thermal radiation that, in the semi-classical description, appears not to preserve the correlations needed to reconstruct the original state. This paper examines how the information paradox arises from this apparent pure-to-mixed state evolution, and uses the Page curve to illustrate the behaviour required if evaporation is ultimately unitary. It also considers whether microscopic black-hole degrees of freedom, or quantum “hair,” could preserve information while remaining invisible to the classical no-hair description. Although no complete microscopic solution is established, the analysis shows that classical hairlessness does not by itself require information loss, and that information may instead be encoded in subtle correlations within the full quantum state of the Hawking radiation.

1 Introduction

Black holes are among the most extreme predictions of general relativity, but they also reveal a deep tension between gravity and quantum mechanics. Classically, a stationary black hole is remarkably simple: under appropriate assumptions, its observable properties are determined by its mass, angular momentum, and electric charge. This idea is commonly expressed through the no-hair principle ([1]), which suggests that the detailed properties of the matter that formed a black hole are no longer visible from the outside. Quantum mechanics presents a very different picture. The evolution of an isolated quantum system is expected to be unitary ([2]), meaning information in its initial state is preserved as it evolves. The problem becomes apparent when Hawking radiation is considered. Hawking showed that black holes should emit radiation with an approximately thermal spectrum and, therefore, gradually lose mass ([3]). If a black hole eventually evaporates completely, an initially pure quantum state appears to evolve into thermal radiation that is described by a mixed state. This would mean information has been genuinely lost, apparently contradicting unitarity. This is the black hole information paradox. The paradox is not simply that black holes have no hair, but that the classical no-hair description, semiclassical Hawking radiation, and unitary quantum mechanics appear difficult to reconcile. This paper investigates the question:

Does the classical, no-hair description of black holes necessarily imply information loss, or could microscopic degrees of freedom and correlations in Hawking radiation preserve unitarity?

To explore this, the paper first examines the no-hair principle and the meaning of information in quantum mechanics. It then introduces black-hole entropy and Hawking radiation before examining how the information paradox arises. Finally, the Page curve will be established, which provides a way of distinguishing information-preserving evaporation from information-losing evaporation.

2 Black Holes and the No-Hair Principle

A black hole is formed when a sufficiently massive object collapses under its own gravity [1], concentrating its mass-energy into a region within a causal boundary. This boundary is the event horizon, in which no future-directed signal can escape to an outside observer. In the simplest case of a black hole (one which is uncharged and non-rotational), this horizon can be found exactly at the Schwarzschild radius, which is given by [1]:

$$r_s = \frac{2GM}{c^2} \quad (1)$$

On a physical level, this means that once matter passes the event horizon, no signal it sends - whether light, matter, or radiation - can be observed by an outside observer.

However, this creates a theoretical complication. Consider two stars with different compositions and histories, both collapsing into black holes of the same mass. Once the black holes settle down, is there a way to tell them apart externally? Conventionally, the answer is no. In traditional general relativity, a stationary black hole is fully described by only three parameters: mass, M , angular momentum, J , and electric charge, Q . No matter how intricate the matter falling into the black hole was, nothing else about the black hole is externally measurable. This is the no-hair theorem.

Stationary black hole solutions in general relativity are completely fixed by M , J and Q , giving rise to four commonly discussed classical solutions [4]:

Black Hole	M	J	Q
Schwarzschild	✓	0	0
Kerr	✓	✓	0
Reissner–Norstrom	✓	0	✓
Kerr–Newman	✓	✓	✓

Table 1: The four classical stationary black hole solutions characterised by mass M , angular momentum J , and electric charge Q .

In this case, “hair” means any extra feature that could distinguish one black hole from another just by its formation history. The no-hair theorem states that there are none.

It is worth being precise about what this does and does not imply. No-hair does not imply that information is destroyed; rather, it states that the detailed information about the black hole’s formation is not encoded in its classical external parameters. Whether that information still exists beyond the horizon is left open. As long as the black hole exists, the apparent loss of information need not constitute a fundamental inconsistency: the information may remain encoded within the black hole interior but be inaccessible to exterior observers, rather than having been irretrievably destroyed.

However, that drastically changes if the black hole does not exist forever. If it can lose mass and eventually evaporate, then whatever became of that information has to go somewhere. It must either be preserved and somehow returned to the universe, or the information is fundamentally lost.

3 Information in Quantum Mechanics and Unitarity

To understand why black holes present a fundamental challenge to modern physics, we must first establish how quantum mechanics preserves information.

A quantum system can be fully described by its state, which is represented as a vector in a Hilbert space. The time evolution of this state is governed by the Schrödinger equation, which is deterministic. This means that if given the state at one time, the state at any later time is fixed entirely. Therefore, the evolution can, in theory, run backwards, resulting in recovery of an earlier state. This evolution is implemented by a unitary operator, $\hat{U}$, satisfying $\hat{U}^\dagger \hat{U} = \hat{I}$. This preserves the norm of the state vector and, therefore, ensures that the total probability of all possible outcomes remains equal to one at all times [2]. This property, unitarity, is not simply a minor technical assumption. It is one of the foundational premises of quantum mechanics, and, therefore, its violation would mean that probabilities are not conserved under time evolution.

A key distinction, however, is the difference between mixed and pure states. A pure state corresponds to complete knowledge of a system and therefore can be written as a single state vector. A mixed state describes a statistical ensemble of possible states and is represented by the density matrix that is unable to be reduced to a single term. Unitarity evolution has a direct consequence for this distinction: since $\hat{U}$ acts oppositely on the full state, a pure state evolving under unitarity dynamics remains pure for any amount of time. It can become entangled with other degrees of freedom, and information within it may become mixed up, but it can never become a mixed state while the evolution remains unitary and the system is closed, or assumed to be closed.

In 1976, Hawking made this precise in the context that would become the information paradox [5]. This frames non-unitary evolution as equivalent to a non-unitary scattering matrix relating an initial state to a final state. This would mean that the final state of the system is not only difficult to predict from the initial data but also is underdetermined by it.

This principle comes into direct tension with black hole physics. If the matter that collapses to form a black hole is treated as a pure quantum state, unitarity requires that this same information must remain fully recoverable from the state of the universe at any later time. This includes after the black hole itself has evaporated away. Whether the radiation Hawking predicted actually satisfies this requirement, or instead settles into a genuinely mixed thermal state, is the question key to the information paradox.

4 Black Hole Entropy

In thermodynamics, entropy is the measure of the number of microscopic configurations consistent with a system's microscopic state. The second law of thermodynamics requires that the total entropy of an isolated system never decreases.

In 1973, Bekenstein identified difficulty with this in black holes. If an object carrying entropy falls across an event horizon, its entropy appears to be removed entirely from the observable universe, since no information about the interior of a black hole is accessible from the outside. Bekenstein noted that this apparent violation of the second law could be resolved by recognising a similarity already present in the mathematics of black hole mechanics: the surface area of a black hole's event horizon, like entropy, is a quantity which never decreases (Bardeen, Carter and Hawking 1973). He proposed that this was not a coincidence but rather reflected a genuine entropy associated with the black hole

itself, defining black hole entropy as a measure of the information about the black hole's interior that is inaccessible to the outside.

This is the Bekenstein-Hawking entropy:

$$S_{BH} = \frac{k_B c^3 A}{4G\hbar} \quad (2)$$

In this equation, A stands for the area of the event horizon. Its dependence on area rather than volume is a significant departure from the extensive scaling of entropy in ordinary matter and is an indication that the information describing a black hole's interior may be encoded on its 2D horizon rather than spread across its interior volume.

In 1973, Bekenstein proposed a general second law of thermodynamics: the sum of ordinary entropy outside a black hole and the black hole's own horizon entropy can never decrease. This law resolves the thermodynamic difficulty, but introduces a new consequence. Standard thermodynamics requires that any system with non-zero entropy also possess a non-zero temperature. However, classical general relativity treats a black hole as a perfect absorber that emits nothing and, therefore, has zero temperature. This directly contradicts the entropy Bekenstein assigned to it.

Initially, Hawking rejected Bekenstein's proposal on this basis, arguing that without an identified physical mechanism by which a black hole could emit energy, the congruence with thermodynamics could only be formal rather than literal. Between 1973 and 1975, this position slowly changed. Applying quantum field theory to the curved spacetime surrounding a black hole and building on the formal thermodynamic framework established by Bardeen and Carter, Hawking found that black holes are not, in fact, perfectly black; rather, they emit a thermal radiation set by their surface gravity.

5 Hawking Radiation

Typically, a black hole should only ever grow; matter and radiation fall in, and nothing gets out. However, in 1974, Stephen Hawking showed this wasn't quite true once quantum effects are taken into account. Black holes can actually emit radiation and slowly lose mass. [6]

A useful heuristic picture begins with quantum fluctuations of fields near the event horizon. Quantum field theory implies that the vacuum is not empty, but instead contains continuous energy and fluctuations. This can be pictured as a pair of particles being produced. However, one falls into the black hole, while the other escapes to a distant observer. In this picture, the inward-falling particle can be associated with negative energy relative to a distant observer, reducing the black hole's total energy and, therefore, its mass. Although this particle-pair picture is only a heuristic explanation rather than a formal derivation, it provides an intuitive picture of Hawking radiation. [7]

Hawking's full calculation, using quantum field theory in curved spacetime, shows that this radiation has a thermal spectrum characterised by the Hawking temperature [3],

$$T_H = \frac{\hbar c^3}{8\pi G M k_b}. \quad (3)$$

The key feature is its dependence on mass:

$$T_H \propto \frac{1}{M}. \quad (4)$$

Thus, the smaller the black hole, the hotter it is, and the more rapidly it emits radiation. This suggests that as a black hole loses mass through Hawking radiation, its temperature will increase, causing the rate of energy loss to increase further. This produces an accelerating evaporation process rather than a steady one. Therefore, the semi-classical prediction suggests that a black hole can eventually become extremely small and hot, although the behaviour of the black hole at this final stage is unknown because quantum-gravitational effects are expected to become important.

Each quantum of radiation carries energy away from the black hole. By Einstein's mass-energy relation,

$$E = mc^2, \quad (5)$$

this loss of energy corresponds to a loss of mass. The approximate evaporation time for an uncharged, non-rotating black hole scales as

$$T_{\text{evap}} \propto m^3, \quad (6)$$

meaning that more massive black holes have extraordinarily long evaporation timescales. For astrophysical black holes, Hawking radiation is therefore expected to be extremely weak and has not yet been directly observed.

This is where the no-hair principle becomes relevant again. Hawking's calculations show that the outgoing radiation depends only on M , J and Q - the black hole's macroscopic properties - not on the detailed makeup of whatever fell in to form it. It looks purely thermal, carrying no distinguishable imprint of the infalling matter's specific quantum state. It would produce the same Hawking temperature regardless. [3]

These two facts now sit side-by-side: black holes are predicted to evaporate away completely, and the radiation they emit while doing so appears completely featureless and memoryless. What that combination means for the fate of the information within the black hole (and why it strikes at something quantum mechanics treats as sacred) is the question the next section takes up directly.[5]

6 How the Information Paradox Arises

The previous sections give us all the pieces necessary for the information paradox. Quantum mechanics requires the evolution of an isolated system to be unitary, meaning that an initially pure quantum state must ultimately evolve into another pure state. At the same time, Hawking's semiclassical calculation predicts that a black hole emits approximately thermal radiation and can eventually evaporate away completely. The paradox arises when these two statements are considered together.

Suppose some matter begins in a pure quantum state,

$$|\psi_{\text{initial}}\rangle. \quad (7)$$

The matter then undergoes gravitational collapse and forms a black hole. The no-hair principle tells us that, after the black hole settles down, its classical exterior is described only by a small number of macroscopic quantities, such as M , J , and Q . The detailed quantum state of the matter that formed the black hole is therefore not represented by any additional classical parameter visible to an outside observer.

This alone does not violate quantum mechanics. While the black hole still exists, the information could, in principle, remain encoded in microscopic degrees of freedom

behind or associated with the event horizon. The difficulty instead appears when Hawking radiation causes the black hole to evaporate.

One way of seeing the problem more precisely is through entanglement. Hawking radiation can be described semiclassically in terms of quantum field modes that extend across the event horizon. In a simplified description of one mode, the quantum state takes an entangled form schematically similar to

$$|\Psi\rangle \propto \sum_{n=0}^{\infty} e^{-\beta\omega n/2} |n\rangle_{\text{out}} |n\rangle_{\text{in}}, \quad (8)$$

where $\beta = 1/(k_B T_H)$, $|n\rangle_{\text{out}}$ describes the outgoing mode accessible to a distant observer, and $|n\rangle_{\text{in}}$ describes its partner associated with the black-hole interior. The complete state containing both sets of modes can therefore remain pure.

However, an outside observer only has access to the outgoing radiation. The interior degrees of freedom must consequently be traced out. If the total state has density matrix

$$\rho_{\text{total}} = |\Psi\rangle\langle\Psi|, \quad (9)$$

then the density matrix describing only the radiation is

$$\rho_{\text{out}} = \text{Tr}_{\text{in}}(\rho_{\text{total}}). \quad (10)$$

Performing this trace gives a thermal reduced density matrix of the approximate form

$$\rho_{\text{out}} = (1 - e^{-\beta\omega}) \sum_{n=0}^{\infty} e^{-\beta\omega n} |n\rangle_{\text{out}}\langle n|. \quad (11)$$

Thus, although the combined interior–exterior system may be in a pure state, the radiation viewed by itself is mixed. There is nothing unusual about this at first. The same occurs for any entangled quantum system when one subsystem is ignored. As long as the black hole remains, the missing correlations could still reside in the degrees of freedom associated with the black hole interior.

The contradiction appears if the black hole evaporates completely. In Hawking’s original semiclassical picture, the evaporation eventually leaves only the outgoing radiation. If that radiation remains described by a thermal mixed state, while the interior degrees of freedom have disappeared with the black hole, there is no remaining subsystem with which the radiation can be entangled. An initially pure state would therefore have evolved according to

$$|\psi_{\text{initial}}\rangle\langle\psi_{\text{initial}}| \longrightarrow \rho_{\text{thermal}}, \quad (12)$$

where

$$\text{Tr}(\rho_{\text{thermal}}^2) < 1. \quad (13)$$

The left-hand side represents a pure state, for which $\text{Tr}(\rho^2) = 1$. The right-hand side represents a mixed state. Ordinary unitary quantum evolution cannot produce such a transformation for a complete closed system [5].

Another way of expressing the same problem is through entropy. A pure state has zero von Neumann entropy,

$$S = -\text{Tr}(\rho \ln \rho) = 0. \quad (14)$$

However, if complete evaporation leaves behind genuinely thermal radiation, the final state has non-zero von Neumann entropy,

$$S_{\text{final}} > 0. \quad (15)$$

Unitary evolution preserves the von Neumann entropy of the complete quantum state. Therefore, a transition from an initially pure state with $S = 0$ to a final mixed state with $S > 0$ would represent a failure of unitarity.

This is stronger than simply saying that the information is extremely difficult to recover. In ordinary thermalisation, information about an initial state may become scrambled into complicated correlations between many particles while still remaining present in the complete quantum state. In the original Hawking calculation, the concern is instead that the final radiation appears not to contain the correlations required to reconstruct the initial state. Many distinct initial configurations could therefore appear to produce the same thermal final description [3, 5].

This gives the information paradox its basic form. General relativity allows matter carrying quantum information to pass through an event horizon. Quantum field theory in curved spacetime predicts Hawking radiation and the eventual evaporation of the black hole. Yet, quantum mechanics requires the complete evolution to remain unitary. When taken at face value, the semiclassical picture appears unable to satisfy all three statements simultaneously.

Therefore, the word “paradox” does not necessarily mean that the information is truly destroyed in nature. Rather, it indicates that the assumptions used in the semiclassical description cannot all remain exact throughout the complete evaporation process. Some additional physics must intervene: information might be encoded in subtle correlations within the Hawking radiation, the semiclassical description of the horizon might fail, new microscopic black hole degrees of freedom might become important, or quantum gravity may change how spacetime and information are described. Determining which of these possibilities restores unitary evolution is the central black hole information problem.

The next section makes this conflict quantitative. If information is ultimately preserved, the entropy of the Hawking radiation cannot simply continue increasing until evaporation is complete. Instead, its entropy must eventually turn over and return toward zero. This expected unitary behaviour is described by the Page curve [8].

7 The Page Curve and Information Recovery

The information paradox becomes much easier to understand by asking a simple question: how does the information content of Hawking radiation change as a black hole evaporates? The Page curve [8] provides a theoretical answer by tracking the entanglement entropy of the radiation over time used by physicists.

Consider a black hole that forms from an initially pure quantum state and then begins to evaporate. At any time, the complete system can be divided into two parts: the radiation that has escaped, R , and the remaining black hole, B . If the complete system evolves unitarily, the combined state can remain pure even though either part, considered separately, appears mixed. The entropy of the radiation is described by its von Neumann entropy (where ρ is the energy density matrix)[9],

$$S_R = -\text{Tr}(\rho_R \ln \rho_R). \quad (16)$$

At the beginning of evaporation, essentially no radiation has escaped (since the ρ has one eigenvalue of 1), so

$$S_R \approx 0. \quad (17)$$

As Hawking radiation is emitted, the radiation becomes entangled with the remaining black hole. If we consider the radiation alone and ignore the degrees of freedom still associated with the black hole, it appears increasingly mixed. Thus, the entropy rises. This initial increase is consistent with Hawking's semiclassical calculation. Early radiation appears approximately thermal and does not obviously reveal the microscopic information contained in the original state.

The crucial difference arises later. If the black hole evaporates completely and the overall evolution is unitary, the final radiation must contain all the information required to describe the original pure state. Since there is eventually no remaining black hole with which the radiation can be entangled, the complete final radiation must itself return to a pure state.

The behaviour we get from a Unitary model is the Page Curve, where radiation entropy increases until half the black hole's evaporation lifetime, a point called the Page Time. At the Page Time, entropy is at its maximum. This means that before this, newly emitted radiation is predominantly entangled with the remaining black hole, causing S_R to increase. After the Page Time, information must increasingly appear in correlations involving the newly emitted radiation and the radiation that escaped earlier. The entropy of the total radiation can therefore decrease, even though the amount of radiation continues to increase.

There is an important point: information recovery does not require each individual Hawking particle to carry an obvious message about the matter that formed the black hole. Information can be encoded in correlations. A good example is that in a jigsaw puzzle, one piece doesn't reveal much about the entire puzzle, but the correlations between many pieces can give us a clearer, complete picture. What the Page curve really does is give us a clear distinction between two different pictures of black-hole evaporation. One picture is where entropy always rises in a semiclassical Hawking view, while the other picture is the unitarity view, where entropy reaches a maximum and turns over after the Page Time to conserve information. Modern theoretical work has provided increasingly strong evidence that a unitary Page curve behaviour can arise when quantum-gravitational effects are properly included. However, the detailed physical mechanism by which information is encoded and recovered remains deeply connected to unresolved questions about the microscopic structure of black holes.

8 Can Black Hole Hair Restore Unitarity?

The classical no-hair principle tells us that, under appropriate assumptions, a stationary black hole can be described externally by only three macroscopic quantities: its mass, M , angular momentum, J , and electric charge, Q . This raises a question as to whether a black hole could possess additional microscopic information that is invisible to the classical description but still influences its quantum evolution.

The answer is not ruled out by the no-hair principle itself. The classical no-hair theorem constrains what can be measured through the classical external fields of a stationary black hole. However, it does not necessarily imply that the complete quantum state is

uniquely specified by (M, J, Q) . Any extra properties we associate with a black hole are called "hairs".

For example, two quantum black holes could, in principle, have identical classical properties while differing in their microscopic states:

$$|\text{BH}_A\rangle = |M, J, Q, \alpha\rangle \quad (18)$$

$$|\text{BH}_B\rangle = |M, J, Q, \beta\rangle \quad (19)$$

where

$$\alpha \neq \beta \quad (20)$$

represents information not visible in the classical parameters. To a classical observer, these black holes would appear identical. Yet, at the quantum level, they could still be fundamentally different states. This distinction provides a possible route around the apparent conflict between no-hair and unitarity.

Suppose the microscopic label α affects extremely subtle correlations in Hawking radiation. The average spectrum observed from the black hole could remain approximately thermal

$$P(E) \approx P_{\text{thermal}}(E) \quad (21)$$

while the complete quantum state of all emitted radiation still depends on α . The information would therefore not necessarily appear in the properties of individual particles. Instead, it could be encoded in the relationships between many particles.

So, two black holes might therefore produce radiation that looks identical when examined one particle at a time:

$$P_A(E) = P_B(E), \quad (22)$$

but differs when correlations are considered:

$$P_A(E_1, E_2, E_3, \dots, E_N) \neq P_B(E_1, E_2, E_3, \dots, E_N) \quad (23)$$

This is the key idea behind the possibility of quantum hair: information can exist in features of the complete quantum state that are absent from the simple classical description. However, this conclusion shouldn't be overstated. Introducing an unspecified microscopic variable, α , does not solve the black-hole information paradox. A successful theory must explain what these degrees of freedom physically are, how they evolve, and how they transfer information into the radiation without contradicting the principles underlying black hole physics. Several modern approaches to quantum gravity attempt to address these questions, but there is not yet a single universally accepted microscopic description of an evaporating black hole. For now, many hypotheses and theories on "hairs" and additional quantum structure remain speculative.

9 Conclusion

The black hole information paradox arises from the apparent incompatibility between three powerful ideas: the classical no-hair description of black holes, Hawking's prediction of approximately thermal radiation, and the unitarity of quantum mechanics. At first, a black hole appears to erase information about the complicated system that formed it, leaving only a small number of macroscopic properties such as M , J , and Q . If the black hole then evaporates completely through purely thermal radiation, the final state

appears unable to contain the information present in the initial quantum state. The no-hair principle describes the classical, externally observable properties of stationary black holes. It does not prove that the complete quantum state contains no additional microscopic degrees of freedom. The Bekenstein–Hawking entropy itself suggests that an enormous number of microscopic states can correspond to the same macroscopic black hole.

The Page curve provides a particularly useful way to express the requirements of unitarity: if evaporation is unitary, the entropy of the radiation should eventually decrease after reaching a maximum, indicating that information is being recovered through correlations in the radiation. This differs from the simple semiclassical picture in which the radiation remains thermal, and its entropy continually increases. An interesting resolution could be that information may not be encoded in the properties of individual Hawking particles, but in subtle correlations between many emissions. In such a scenario, the radiation could appear approximately thermal when examined locally, while still containing the information required for the overall evolution to remain unitary. This provides a conceptual way in which a black hole could remain classically “hairless” while possessing a much richer quantum structure. Nevertheless, this should not be interpreted as a solution to the information paradox. A complete explanation would need to identify the microscopic degrees of freedom responsible for the black hole entropy and demonstrate, within a consistent theory of quantum gravity, how they transfer information into the outgoing radiation. The apparent paradox instead arises from treating the semiclassical thermal description of Hawking radiation as the complete quantum description of the evaporation process. If information is preserved, it must be encoded in aspects of the radiation or the black hole state that are invisible to the simplest classical and semiclassical descriptions. Black holes provide more than an unusual prediction of general relativity. They expose fundamental questions about the relationship between spacetime, gravity, quantum mechanics, and information.

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Fly-By-Wire Systems

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1. Abstract

This paper discusses the usage and evolution of aeroplane control systems; more specifically, the fly-by-wire system (FBW). As aircraft developed, ever since the Wright brothers reached the skies in 1903, mechanical control systems were traditionally used. Controls were linked to mechanical systems via cables and pistons, where the input of the pilot directly affected the output exhibited by the aeroplane. Currently, most aeroplanes use FBW systems, where the input of the user is carried to computers via cables, which decide which output to give actuators (an actuator is a mechanical or electromechanical device that converts an energy source and a control signal into physical movement). When a pilot moves the control column, a signal is sent to a computer through multiple wires or channels.

The computer receives the signals, which are then sent to the control surface actuator, resulting in surface motion. The potentiometers in the actuator send a signal back to the computer reporting the position of the actuator. When the actuator reaches the desired position, the two signals (incoming and outgoing) cancel each other out, and the actuator stops moving.

The use of FBW systems has been a delight of the aircraft industry. It reduces aircraft weight, meaning reduced fuel consumption and ultimately lower cost or longer maximum flight duration [1].

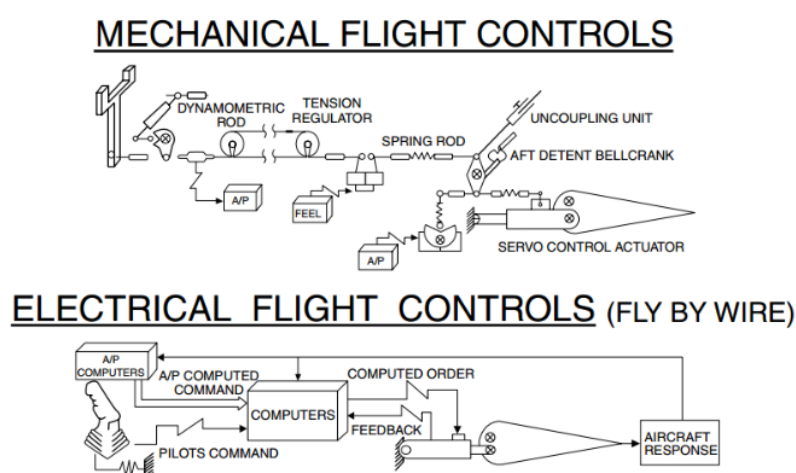


Figure 1: A graphical overview of the FBW system, compared to the prior mechanical flight control system. It is evident that the FBW control system reduced the need for a lot of heavy and expensive hardware.

2. Flight Control Systems

FBW systems employ electronic control signals and flight-control computers to regulate an aircraft's response to pilot commands. The system continuously processes information obtained from onboard sensors, including measurements of airspeed, altitude, acceleration, and angle of attack, to determine the appropriate response of the aircraft.

Aircraft motion is controlled about three principal axes - roll, pitch, and yaw - with corresponding control surfaces producing the aerodynamic moments required to alter

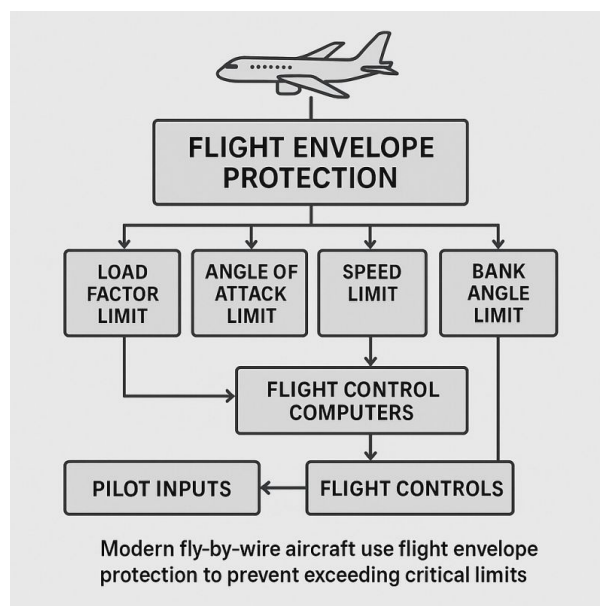


Figure 2: A visual interpretation of the flight envelope system [4].

the aircraft's orientation. The flight control computer interprets flight control laws to determine the appropriate control response. [2]

An important function of these systems is flight envelope protection, which can prevent the aircraft from exceeding critical aerodynamic or structural limits. For example, if a pilot commands a large nose-up pitch input and the aircraft approaches its stall angle of attack, the flight control system can limit the commanded elevator deflection to prevent the aircraft from entering an unsafe aerodynamic condition [3]. Consequently, FBW technology provides continuous and automated regulation of aircraft response, improving stability and allowing the aircraft to remain within its defined operating envelope.

2.1 How Does It Work?

A pilot input, such as movement of the control column or sidestick, generates an electrical signal that is transmitted to the flight control computers. These computers interpret the pilot's command and determine the appropriate aircraft response based on the current flight conditions and applicable control laws. They calculate the required deflection of the elevators, ailerons, rudder, and other flight control surfaces to produce the intended manoeuvre while maintaining aircraft stability [5].

The resulting commands are transmitted to the corresponding control-surface actuators, which convert the electrical commands into the physical movement of the control surfaces. Feedback sensors continuously measure the position of these surfaces and transmit this information back to the flight control computers, allowing any deviations from the commanded position to be detected and corrected [6].

To ensure reliability and fault tolerance, FBW systems typically incorporate multiple flight control computers and independent signal channels, allowing the aircraft to remain controllable in the event of a component or channel failure. Digital data buses transmit signals between the flight-control computers, sensors, and other aircraft systems, while electro-hydraulic or electrically-powered actuators provide the force re-

quired to move the control surfaces. This continuous cycle of command, actuation, and feedback enables precise and responsive control of the aircraft throughout flight.

2.2 Aerodynamic Equations

Traditional flight relies on direct mechanical translation of force from a pilots hand control surfaces. When a surface deflects, the angle of attack on the wing or tail will change. This creates a different pressure distribution across the top and bottom surfaces, generating an aerodynamic lift force that pitches, rolls, or yaws the aircraft:

$$L = C_L \frac{1}{2} \rho V^2 S$$

- L - Lift force
- C_L - Lift coefficient
- ρ - Air density
- V - True airspeed
- S - Wing area

[7]

2.3 Processing Electrical Systems

In a FBW system, transducers measure the displacement or force applied to the sidestick. The computer calculates the ideal aerodynamic response, such as dynamic pressure, $\frac{1}{2} \rho v^2$, so that a minor input does not overstress the airframe.

A dynamic system is influenced by aerodynamic forces, thrust, and gravity. FBW computers process electrical signals sent from sensors to control this system. This physical behaviour can be modelled using a state-space representation, expressed in matrix form as:

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned} \tag{1}$$

- $\dot{x}(t)$ - State vector
- $u(t)$ - Control vector
- A - System matrix
- B - Input matrix

[8]

This engineering system is described by differential equations that relate an input $u(t)$ to an output $y(t)$. The output of the system depends not only on the input, but also on derivatives of the output itself.

The system can be written as

$$L(y) = b_2\ddot{y}(t) + b_1\dot{y}(t) + b_0y(t), \quad (2)$$

where b_0 , b_1 and b_2 are constant coefficients. If the input to the system is $u(t)$, the differential equation is:

$$b_2\ddot{y}(t) + b_1\dot{y}(t) + b_0y(t) = u(t). \quad (3)$$

This can be expressed as:

$$L(y) = u. \quad (4)$$

The objective is to determine the output y as a function of the input u . Therefore, the inverse operator is introduced, such that

$$y = G(u) = L^{-1}(u). \quad (5)$$

Here, L^{-1} represents the operation required to solve the differential equation for $y(t)$ given the input $u(t)$. The state-space representation provides a systematic method for performing this transformation.

Starting from the second-order differential equation,

$$b_2\ddot{y}(t) + b_1\dot{y}(t) + b_0y(t) = u(t), \quad (6)$$

the equation can first be rearranged so that the highest derivative is expressed in terms of the remaining variables:

$$\ddot{y}(t) = -\frac{b_1}{b_2}\dot{y}(t) - \frac{b_0}{b_2}y(t) + \frac{1}{b_2}u(t). \quad (7)$$

We now introduce two state variables. The first state variable is chosen to represent the output itself,

$$x_1(t) = y(t), \quad (8)$$

while the second state variable represents its first derivative,

$$x_2(t) = \dot{y}(t). \quad (9)$$

Differentiating the first state variable gives:

$$\dot{x}_1(t) = \dot{y}(t) = x_2(t). \quad (10)$$

Similarly, differentiating the second state variable gives:

$$\dot{x}_2(t) = \ddot{y}(t). \quad (11)$$

Substituting the differential equation into this expression gives:

$$\dot{x}_2(t) = -\frac{b_1}{b_2}x_2(t) - \frac{b_0}{b_2}x_1(t) + \frac{1}{b_2}u(t). \quad (12)$$

The two first-order equations can, therefore, be written together as:

$$\begin{aligned} \dot{x}_1(t) &= x_2(t), \\ \dot{x}_2(t) &= -\frac{b_0}{b_2}x_1(t) - \frac{b_1}{b_2}x_2(t) + \frac{1}{b_2}u(t). \end{aligned} \quad (13)$$

These equations can be expressed compactly using matrix-vector notation:

$$\frac{d}{dt} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{b_0}{b_2} & -\frac{b_1}{b_2} \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{b_2} \end{pmatrix} u(t). \quad (14)$$

By defining the state vector as

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}, \quad (15)$$

the system can be written in the standard state-space form

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + Bu(t) \quad (16)$$

where

$$A = \begin{pmatrix} 0 & 1 \\ -\frac{b_0}{b_2} & -\frac{b_1}{b_2} \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \frac{1}{b_2} \end{pmatrix}. \quad (17)$$

Since the output is defined by $y(t) = x_1(t)$, it can be written as

$$y(t) = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}. \quad (18)$$

Therefore,

$$y(t) = C\mathbf{x}(t) \quad (19)$$

with

$$C = \begin{pmatrix} 1 & 0 \end{pmatrix}. \quad (20)$$

The same procedure can be generalised to an n th-order linear time-invariant (LTI) dynamical system. Instead of representing the system using a single higher-order differential equation, the system is represented using a set of coupled first-order differential equations.

The general continuous-time state-space representation is

$$\begin{aligned} \dot{\mathbf{x}}(t) &= A\mathbf{x}(t) + B\mathbf{u}(t), \\ \mathbf{y}(t) &= C\mathbf{x}(t) + D\mathbf{u}(t). \end{aligned} \quad (21)$$

As seen before

$$\mathbf{x}(t) \in \mathbb{R}^n, \quad \mathbf{y}(t) \in \mathbb{R}^m, \quad \mathbf{u}(t) \in \mathbb{R}^p. \quad (22)$$

The matrices A , B , C , and D describe the dynamics, input, output, and direct input-output relationships of the system, respectively. Their dimensions are:

$$A \in \mathbb{R}^{n \times n}, \quad B \in \mathbb{R}^{n \times p}, \quad (23)$$

$$C \in \mathbb{R}^{m \times n}, \quad D \in \mathbb{R}^{m \times p}. \quad (24)$$

[9]

Thus, the state-space formulation transforms an n th-order differential equation into a system of n coupled first-order equations. This mathematical framework is implemented within the flight control computer to process digitised sensor inputs. This system replaces traditional mechanical cables, providing automatic protection to prevent dangerous manoeuvres.

3. Discussion

The FBW system offers many advantages over traditional analogue systems. These include:

- Reduced weight.
- Improved reliability.
- Greater damage tolerance.
- More effective control of highly sensitive, manoeuvrable aircraft.
- Improved pilot and passenger safety through protection software.
- Reduced effects of air disturbance, resulting in lower fatigue loads and improved passenger comfort.
- Reduced maintenance requirements.
- Reduced pilot training costs due to greater operational similarity across an aircraft family.
- Improved fuel efficiency due to the system's lower weight and reduced complexity compared with manual systems.
- Reduced pilot workload, as flight control computers continuously manage aspects of the aircraft's flight.

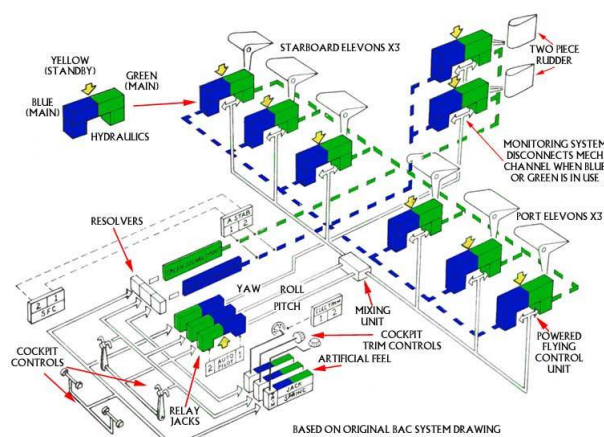


Figure 3: The FBW system has an intricate electrical framework that reduces the need for heavy machinery, reducing loads and, therefore, fuel consumption, and ultimately reducing costs and increasing efficiency[10].

The FBW system uses feedback control parameters - including airspeed, attitude, angle of attack, and Mach number - to ensure the aircraft stays within its designed flight envelope. The Airbus approach features a flight envelope with strict restrictions; unless the pilot chooses direct law, the control laws have complete command control. The pilot has the last say over how the aircraft operates because the Boeing strategy includes a flight envelope with adjustable constraints [11].



Figure 4: An example of a gyroscope used on aircraft. The gyroscope reflects physical changes in orientation and/or angular velocity and reports this to the aircraft's computers[1].

FBW control systems (FBWCS) are a great step up from the traditional analogue setup, for the aforementioned reasons. This is evident when the scale of aircraft world-wide that use digital FBWCS is analysed. Tens of thousands of individual aircraft world-wide utilise digital FBWCS, as it has become the industry standard for nearly all commercial airliners (e.g. Airbus A320, A330, A340, A350 and A380) [12], military fighter jets (e.g. Lockheed Martin F-35 Lightning II, F-22, Rafale) [13], and high-end business aircraft (e.g. Dassault Falcon 7X, 8X, 10X and Gulfstream G500/G600) [14].

However, there is a new developing flight control system that is a further improvement on the revolutionary FBWCS. The FBL (Fly-by-light) control system utilises fibre-optic cables instead of electrical cables to transmit data in the form of visible light, rather than as electrical signals. As we are aware, photons travel faster than electrons (as electrons have mass, so they can never travel at the speed of light). Therefore, the FBL control system has the ability to transmit data at a faster pace than FBWCS, resulting in a more efficient control system. Combined with reduced weight and cable management, FBL control systems have the same advantages over FBWCS that FBWCS has over traditional analogue control systems. An increasing number of aircraft companies are starting to adopt the new FBL control system, such as Kawasaki (P-1 and C-2) [15] and the German Aerospace Centre (Eurocopter EC135 test helicopter) [16].

This evidence would suggest that FBL control systems are a suitable improvement from FBWCS.

4. Conclusions

The FBWCS has provided the aerospace industry with substantial advantages. These include a reduction in total system weight, the development of multiple redundant control paths, and significantly increased passenger safety across all aircraft using FBWCS. The transition from analogue to digital FBWCS represented a significant advancement in the evolutionary process of this technology. The current generation of aircraft utilise FBWCS.

However, the next generation of aircraft will be designed around the use of FBL flight control systems. FBL offers many benefits, including reduced weight, high bandwidth, compact sizing, and resistance to electromagnetic interference and high-intensity radiated fields.

Therefore, although the FBWCS represented a significant advancement in flight control technology, it is anticipated that FBL will form the basis of the next generation of flight control systems. FBL offers enhanced protection against the increasingly demanding operating conditions encountered in military aviation. Its inherent characteristics are the driving factors behind the technological advancements required to develop FBL into a viable alternative to existing aircraft flight control technologies. The use of optical fibres within aviation applications using FBL may help address current flight control issues and provide answers to questions related to aircraft stability.

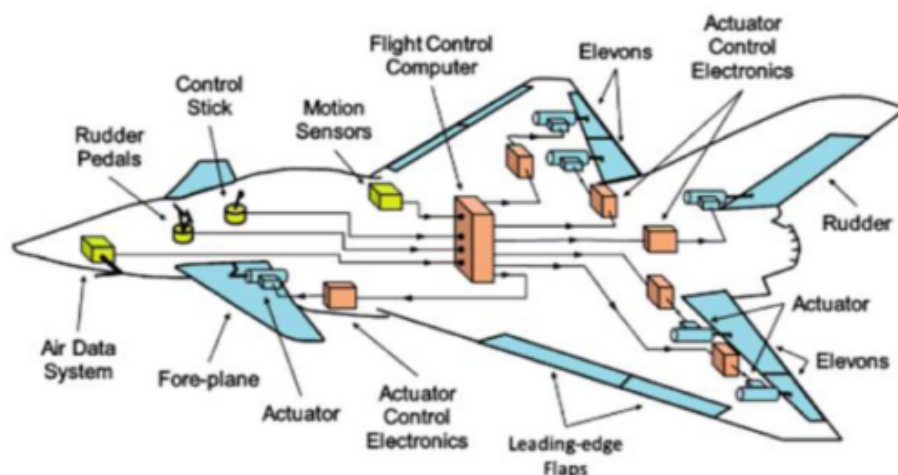
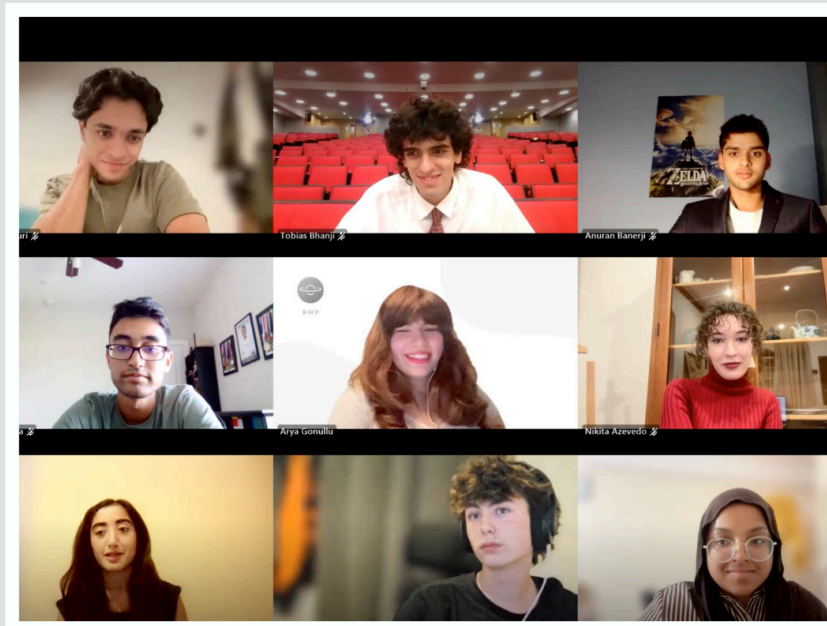


Figure 5: The FBL system carries a similar framework to the FBW system, but uses optical fibres in order to reduce weight and increase bandwidth[17].

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SECTION 04

HAWKING RADIATION

Vacuum fluctuations near the horizon create particle pairs. One partner escapes as real radiation; the other falls in carrying negative energy — the black hole loses mass to balance the books.

$$T_H = \hbar c^3 / (8\pi G M k_B)$$

Smaller mass → hotter → faster evaporation. The process runs away.

$$T_H \propto 1/M$$

Radiating energy away is, directly, losing mass.

Particle pairs at the horizon

Pros and cons

Strengths	Limitations
Characteristically explains existence dark matter	But many characteristics don't match
Explains inflation (to an extent)	Heavy particle that decays rapidly
Falsifiable	Could invoke fifth forces

Introduction

This paper will discuss the use of airplane control systems; more specifically, the fly-by-wire control system (FBWCS).

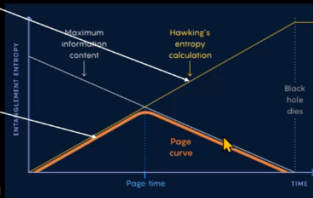
- Traditionally, airplanes would use manually controlled control systems.
- They now use digitally controlled control systems, where the input of the user is sent as an electrical signal to the computer.
- The input of the user does not directly control the physical output of the actuators.



SECTION 06

THE PAGE CURVE THEORY & INFORMATION RECOVERY

- Under classical assumptions, we'd expect entropy to keep rising as the black hole evaporates through Hawking Radiation. This doesn't take into account any quantum mechanics or any hair.
- The Page Curve is a model of how entropy changes in a black hole under Unitarity
- Von Neumann entropy follows:
 $S(\rho) = -\text{Tr}(\rho \ln(\rho))$
- These are two different models that show the exact same kind of Hawking Radiation emission from a black hole, one governed by Hawking's entropy calculation and one governed by Unitarity's necessity to keep information restored and traceable.



Charlie Davies (GUTUYA2 MBQ)

SUMMARY

CONCLUSIONS

Summary:

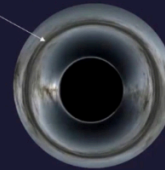
Our investigation shows the information paradox arises from the interactions of two principles, rather than a single cause. The no hair theorem states a stationary black hole is fully described externally by just mass, spin and charge meaning nothing else about infalling matter survives classically. This collides with unitarity which is the quantum requirement stating that evolution stays reversible and pure states remain pure. Hawking radiation is the bridge between the two as it states that particle pairs form at the horizon, one escapes while the other falls in with negative energy. This causes mass loss until the black hole evaporates. Because no hair restricts what this radiation can depend on, it emerges as purely thermal so complete evaporation converts an initial pure state into a mixed one thus violating unitarity.

We don't actually know what dark matter is but we do know it exists through different effects it has like gravitational lensing on massive galaxies or rotation speed of galaxies.

About 27% of all mass-energy content is thought to be dark matter

About 85% of all matter content is thought to be dark matter

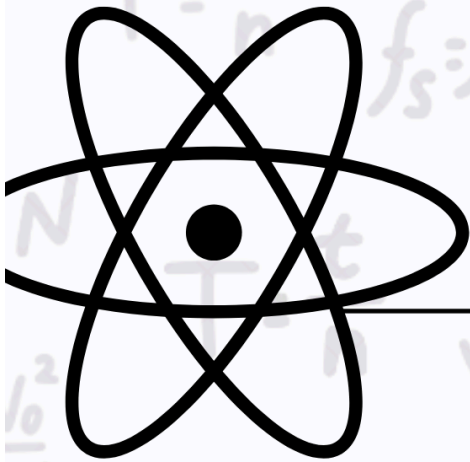
Gravitational lensing



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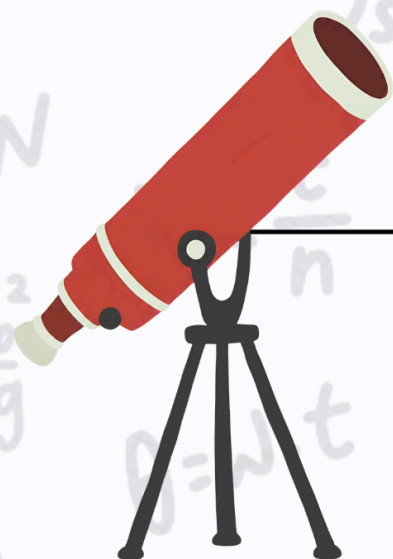


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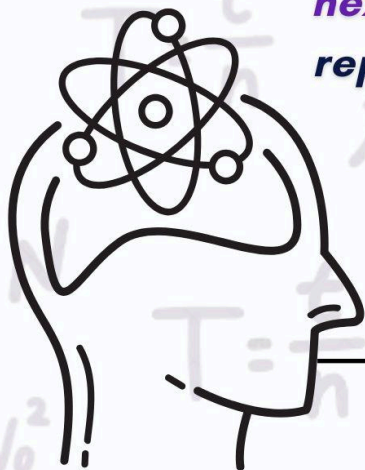
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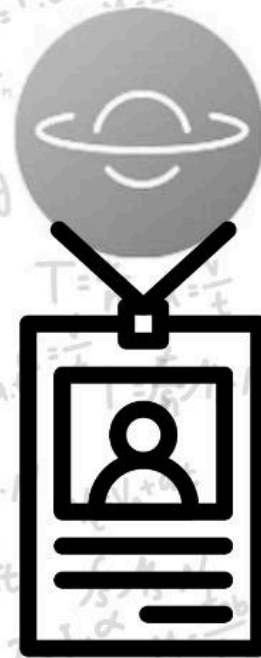
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If you've made it this far, I hope you've found something in this issue that made you stop, think, or perhaps sent you down a scientific rabbit hole of your own.

This has been my first issue as editor of *The Starry Messenger*, and while I'll admit I've found it slightly daunting, I've really enjoyed seeing it come together. I've had the opportunity to dive into topics outside my usual field, and I've learned plenty along the way.

I'd like to say a genuine "thank you" to all our contributors and everyone who has helped put this issue together. It really is a team effort!

And don't forget: if you're in Year 13, an undergraduate, or a graduate and would like your work featured here, we're accepting submissions for December's issue until 30 November. Email arya.gsyp@gmail.com to ask any questions or to submit your work.

I look forward to seeing what arrives for the next issue!

Hayley
Editor, *The Starry Messenger*



Arya Lal Gonullu, Chief Executive of GSYF, named the science magazine after Galileo Galilei's groundbreaking work, *Sidereus Nuncius* (The Starry Messenger). In this historic book, Galileo had unveiled three revolutionary discoveries he made with it: Sunspots, the Moons of Jupiter, and that the Moon is not a perfect sphere. Gonullu hopes that physics will soon experience a new “Starry Messenger” breakthrough through the new rising stars of the field.

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